# RADIO TEST REPORT

| | | |
|---|---|---|
| **Applicant** | : | Shenzhen HOPE Microelectronics Co., Ltd |
| **Address of Applicant** | : | 30/F, Block A, Building 8, Phase 3,Vanke Cloud City,Xili Sub-district,Nanshan,Shenzhen,GD,P.R.China |
| **Manufacturer** | : | Shenzhen HOPE Microelectronics Co., Ltd |
| **Address of Manufacturer** | : | 30/F, Block A, Building 8, Phase 3,Vanke Cloud City,Xili Sub-district,Nanshan,Shenzhen,GD,P.R.China |
| **Equipment under Test** | : | LoRa module |
| **Model No.** | : | RFM90C |
| **Test Standard(s)** | : | ETSI EN 300 220-1 V3.1.1(2017-02)<br>ETSI EN 300 220-2 V3.2.1 (2018-06) |
| **Report No.** | : | DDT-RE25091517-1E01 |
| **Issue Date** | : | 2025/12/30 |
| **Issued By** | : | Guangdong Dongdian Testing Service Co., Ltd.<br>Unit 2, Building 1, No. 17, Zongbu 2nd Road, Songshan Lake Park, Dongguan, Guangdong, China, 523808 |

REPORT

# Table of Contents

# Test Report Declare

| | | |
|---|---|---|
| **Applicant** | : | Shenzhen HOPE Microelectronics Co., Ltd |
| **Address of Applicant** | : | 30/F, Block A, Building 8, Phase 3,Vanke Cloud City,Xili Sub-district,Nanshan,Shenzhen,GD,P.R.China |
| **Equipment under Test** | : | LoRa module |
| **Model No.** | : | RFM90C |
| **Manufacturer** | : | Shenzhen HOPE Microelectronics Co., Ltd |
| **Address of Manufacturer** | : | 30/F, Block A, Building 8, Phase 3,Vanke Cloud City,Xili Sub-district,Nanshan,Shenzhen,GD,P.R.China |

**Test Standard Used:**

ETSI EN 300 220-1 V3.1.1(2017-02), ETSI EN 300 220-2 V3.2.1 (2018-06)

**We Declare:**

The equipment described above is tested by Guangdong Dongdian Testing Service Co., Ltd. and in the configuration tested the equipment complied with the standards specified above. The test results are contained in this test report and Guangdong Dongdian Testing Service Co., Ltd. is assumed of full responsibility for the accuracy and completeness of these tests.

| | | | |
|---|---|---|---|
| **Report No.:** | DDT-RE25091517-1E01 | | |
| **Date of Receipt:** | 2025/11/24 | **Date of Test:** | 2025/11/24 - 2025/12/16 |

| Created: Jacky Huang | Reviewed: Johnson Huang | Approved: Ella Gong |
|---|---|---|
| Jacky Huang | Johnson Huang | Ella Gong<br>GUANGDONG DONGDIAN TESTING SERVICE CO.,LTD.<br>广东东电检测技术有限公司<br>Certificate and Report Seal<br>检验检测专用章 |
| 2025/12/06 | 2025/12/30 | 2025/12/30 |

Note: This report applies to above tested sample only. This report shall not be reproduced in parts without written approval of Guangdong Dongdian Testing Service Co., Ltd.

# Revision History

| Version | Revision Content | Issue Date | Approved |
|---|---|---|---|
| V0 | Initial issue | 2025/12/30 | Ella Gong |
| | | | |

# 1. Summary of Test Results

| No. | Test Parameter | Clause No. | Condition | Result |
|---|---|---|---|---|
| 1 | Operating frequency | 4.2.1 | U | Pass |
| 2 | Unwanted emissions in the spurious domain | 4.2.2 | U | Pass |
| 3 | TX effective radiated power | 4.3.1 | U | Pass |
| 4 | TX Maximum e.r.p. spectral density | 4.3.2 | Applies to EUT using annex B bands I, L. Applies to EUT using DSSS or wideband techniques other than FHSS modulation, using annex C band X. | N/A |
| 5 | TX Duty cycle | 4.3.3 | Not applicable to EUT with polite spectrum access where permitted in annex B, table B.1 or annex C, table C.1 or any NRI | Pass |
| 6 | TX Occupied bandwidth | 4.3.4 | U | Pass |
| 7 | TX out of band emissions | 4.3.5 | Applies to EUT with OCW > 25 kHz. | Pass |
| 8 | TX Transient | 4.3.6 | U | Pass |
| 9 | TX Adjacent channel power | 4.3.7 | Applies to EUT with OCW ≤ 25 kHz. | N/A |
| 10 | TX behaviour under low voltage conditions | 4.3.8 | Applies to battery powered EUT. | Pass |
| 11 | TX Adaptive power control | 4.3.9 | Applies to EUT with adaptive power control using annex C band AA. | N/A |
| 12 | TX FHSS | 4.3.10 | Applies to FHSS EUT using the band 863 MHz to 870 MHz. | N/A |
| 13 | TX Short term behaviour | 4.3.11 | Applies to EUT using annex C bands Y, Z, AA, AB, AC, AD. | N/A |
| 14 | RX sensitivity | 4.4.1 | Applies to EUT with polite spectrum access. | N/A |
| 15 | RX Blocking | 4.4.2 | U | Pass |
| 16 | Clear channel assessment threshold | 4.5.2 | Applies to EUT with polite spectrum access. | N/A |
| 17 | Polite spectrum access timing parameters | 4.5.3 | Applies to EUT with polite spectrum access. | N/A |
| 18 | Adaptive Frequency Agility | 4.5.4 | Applies to EUT with AFA. | N/A |
| Note: N/A is an abbreviation for Not Applicable, and means this item is not applicable for this device or no need to test according to standard. | | | | |

# 2. General Test Information

## 2.1. Description of EUT

| | | |
|---|---|---|
| EUT Name | : | LoRa module |
| Model Number | : | RFM90C |
| EUT Function Description | : | Please reference user manual of this device |
| Power Supply | : | DC 1.8~3.7V |
| Hardware Version | : | V1.2 |
| Software Version | : | V1.0 |
| Antenna Type | : | SMA antenna |
| Max Antenna Gain (dBi) | : | 3 |

| | | |
|---|---|---|
| Radio Technology | : | SRD |
| Operation frequency | : | 868.1MHz, 868.3MHz, 868.5MHz, 864.1MHz, 864.3MHz,864.5MHz |
| Modulation | : | CSS |

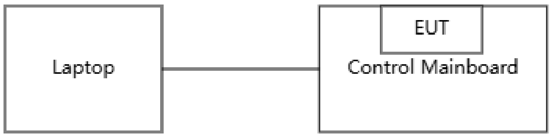
| Channel information | |
|---|---|
| Channel | Frequency (MHz) |
| 0 | 864.1 |
| 1 | 864.3 |
| 2 | 864.5 |
| 3 | 868.1 |
| 4 | 868.3 |
| 5 | 868.5 |

Note: The above EUT information is declared by manufacturer and for more detailed features description please refer to the manufacturer's specifications or User's Manual. The above Antenna information is declared by manufacturer and for more detailed features description please refer to the manufacturer's specifications, the laboratory shall not be held responsible.

## 2.2. Accessories of EUT

| Accessories | Manufacturer | Model number | Description |
|---|---|---|---|
| / | / | / | / |

## 2.3. Block diagram of EUT configuration for test




## 2.4. Decision of final test mode

According pre-test, the worst test modes were reported as below:

| Tested mode, channel, information | | |
|---|---|---|
| Mode | Channel | Frequency (MHz) |
| TX | 0 | 864.1 |
| | 5 | 868.5 |
| RX | 0 | 864.1 |
| | 5 | 868.5 |

## 2.5. Deviations of test standard

No deviation.

## 2.6. Test environment conditions

During the measurement the environmental conditions were within the listed ranges:

| / | Normal Conditions | Extreme Conditions |
|---|---|---|
| Temperature range | +15℃ to +35℃ | LT: -20℃ to HT: +55℃ |
| Humidity range | 20-75% | N/A |
| Power supply | NV DC 3.3V | LV: DC 2.97V to HV: DC 4.29V |
| Note: The Extreme temperature range and extreme voltages are declared by the manufacturer and definitions of abbreviations:<br>NV: Normal Voltage,<br>LV: Low Extreme Test Voltage,<br>HV: High Extreme Test Voltage,<br>NT: Normal Temperature,<br>LT: Low Extreme Test Temperature,<br>HT: High Extreme Test Temperature. | | |

Note: The specific temperature and humidity information of each test item refers to the temperature and humidity record in the corresponding test data.

## 2.7. Test laboratory

Guangdong Dongdian Testing Service Co., Ltd.

Add.: Unit 2, Building 1, No. 17, Zongbu 2nd Road, Songshan Lake Park, Dongguan, Guangdong, China, 523808.

Tel.: +86-0769-38826678, http://www.dgddt.com, Email: ddt@dgddt.com.

CNAS Accreditation No. L6451; A2LA Accreditation Number: 3870.01

FCC Designation Number: CN1182, Test Firm Registration Number: 540522

Innovation, Science and Economic Development Canada Site Registration Number: 10288A

Conformity Assessment Body identifier: CN0048

VCCI facility registration number: C-20087, T-20088, R-20123, R-20240, G-20118

## 2.8. Measurement uncertainty

| Test Item | Uncertainty |
|---|---|
| Bandwidth | 1.1% |
| Peak Output Power (Conducted) (Spectrum analyzer) | 0.86 dB (10 MHz ≤ f < 3.6 GHz); |
| | 1.38 dB (3.6 GHz ≤ f < 8 GHz) |
| Peak Output Power (Conducted) (Power Sensor) | 0.74 dB |
| Power Spectral Density | 0.74 dB (10 MHz ≤ f < 3.6 GHz); |
| | 1.38 dB (3.6 GHz ≤ f < 8 GHz) |
| Frequencies Stability | 6.7 x $10^{-8}$ (Antenna couple method) |
| | 5.5 x $10^{-8}$ (Conducted method) |
| Conducted spurious emissions | 0.86 dB (10 MHz ≤ f < 3.6 GHz); |
| | 1.40 dB (3.6 GHz ≤ f < 8 GHz) |
| | 1.66 dB (8 GHz ≤ f < 26.5 GHz) |
| Uncertainty for radio frequency (RBW < 20 kHz) | $3\times10^{-8}$ |
| Temperature | 0.4 ℃ |
| Humidity | 2 % |
| Uncertainty for Radiation Emission test (9 kHz – 30 MHz) | 3.44 dB |
| Uncertainty for Radiation Emission test (30 MHz - 1 GHz) | 4.70 dB (Antenna Polarize: V) |
| | 4.84 dB (Antenna Polarize: H) |
| Uncertainty for Radiation Emission test (1 GHz - 40 GHz) | 4.10 dB (1 - 6 GHz) |
| | 4.40 dB (6 GHz - 18 GHz) |
| | 3.54 dB (18 GHz - 26 GHz) |
| | 4.30 dB (26 GHz - 40 GHz) |
| Uncertainty for Power line conduction emission test | 3.34dB (150KHz-30MHz) |
| | 3.72dB (9KHz-150KHz) |
| Note: This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2. | |

# 3. Equipment Used During Conductive Test

| Equipment | Manufacturer | Model No. | Serial Number | Due Date | Cal. Interval |
|---|---|---|---|---|---|
| ☐ RF Connected Test (RF Measurement System 1#) | | | | | |
| SIGNAL ANALYZER | R&S | FSQ26 | 101272 | 2026/03/28 | 1 Year |
| Wideband Radio Communication Tester | R&S | CMW500 | 120259 | 2026/03/28 | 1 Year |
| MXG Vector Signal Generator | KEYSIGHT | N5182B | MY59100192 | 2026/03/28 | 1 Year |
| MXG Vector Signal Generator | Agilent | N5182A | MY19060405 | 2026/03/28 | 1 Year |
| RF Control Unit | Tonscend | JS0806-2 | 158060010 | 2026/03/28 | 1 Year |
| TEMP&HUMI Programmable Chamber | ZHIXIANG | ZXGDJS-150L | ZX170110-A | 2026/03/28 | 1 Year |
| Test Software | Tonscend | JS1120-3 | V3.6.21 | N/A | N/A |
| ☒ RF Connected Test (RF Measurement System 2#) | | | | | |
| SPECTRUM ANALYZER | R&S | FSU26 | 201124 | 2026/07/06 | 1 Year |
| Power Sensor | R&S | NRP-Z22 | 101254 | 2026/07/06 | 1 Year |
| Test Software | Tonscend | JS1120-3 | V3.6.21 | N/A | N/A |
| ☐ RF Connected Test (RF Measurement System 3#) | | | | | |
| SIGNAL ANALYZER | R&S | FSV40 | 101407 | 2026/07/06 | 1 Year |
| Wideband Radio Communication Tester | R&S | CMW500 | 117491 | 2026/03/28 | 1 Year |
| EXG Analog Signal Generator | KEYSIGHT | N5173B | MY62153058 | 2026/07/06 | 1 Year |
| MXG Vector Signal Generator | Agilent | N5182A | MY48180912 | 2026/03/28 | 1 Year |
| RF Control Unit | Tonscend | JS0806-2 | 20C8060230 | 2026/03/28 | 1 Year |
| TEMP&HUMI Programmable Chamber | ZHIXIANG | ZXGDJS-150L | ZX170110-A | 2026/03/28 | 1 Year |
| Test Software | Tonscend | JS1120-3 | V3.6.21 | N/A | N/A |
| ☐ RF Connected Test (RF Measurement System 4#) | | | | | |
| Signal &Spectrum Analyzer | R&S | FSV3044 | 101173 | 2026/03/28 | 1 Year |
| Wideband Radio Communication Tester | R&S | CMW500 | 168801 | 2026/03/28 | 1 Year |
| MXG Vector Signal Generator | Agilent | N5182A | MY48180737 | 2026/03/28 | 1 Year |
| PSG Vector Signal Generator | Agilent | E8267D | US49060192 | 2026/08/10 | 1 Year |
| RF Control Unit | Tonscend | JS0806-2 | 21I8060485 | 2026/03/28 | 1 Year |
| TEMP&HUMI Programmable Chamber | ZHIXIANG | ZXGDJS-150L | ZX170110-A | 2026/03/28 | 1 Year |
| Test Software | Tonscend | JS1120-3 | V3.6.21 | N/A | N/A |

# 4. Operating Frequency

## 4.1. Limits

The manufacturer may declare either one or more operating frequencies and operating channels. Operating channel(s) shall be entirely within operational frequency bands allowed by annex B or any NRI. According to EN 300 220-2 Annex B, the EUT shall be using bands K and M.

**Table B.1: EU wide harmonised national radio interfaces from 25 MHz to 1 000 MHz**

| Operational Frequency Band | | Maximum effective radiated power, e.r.p. | Channel access and occupation rules (e.g. Duty cycle or LBT + AFA) | Maximum occupied bandwidth | Other usage restrictions | Band number from EC Decision 2017/1483 /EU [2] |
|---|---|---|---|---|---|---|
| A | 26,957 MHz to 27,283 MHz | 10 mW e.r.p. | No requirement | The whole band | | 28 |
| B | 26,995 MHz,27,045 MHz, 27,095 MHz, 7,145 MHz, 27,195 MHz | 100 mW e.r.p. | ≤ 0,1 % duty cycle | 10 kHz | Model control devices may operate without duty cycle restrictions. | 29, 30, 31, 32, 33 |
| C | 40,660 MHZ to 40,700 MHz | 10 mW e.r.p. | No requirement | The whole band | Video applications excluded. | 35 |
| D | 169,400 MHz to 169,475 MHz | 500 mW e.r.p. | ≤ 1,0 % duty cycle For metering devices duty cycle limit is 10 % | 50 kHz | | 37c |
| E | 169,4000 MHz to 169,4875 MHz | 10 mW | ≤ 0,1 % duty | The whole band | | 38 |
| F | 169,4875 MHz to 169,5875 MHz | 10 mW | ≤ 0,001 % duty cycle Between 00.00 and 06.00 local time a duty cycle limit of 0,1 % may be used | The whole band | Equipment that concentrates or multiplexes individual equipment is excluded. | 39b |
| G | 169,5875 MHz to 169,8125 MHz | 10 mW | ≤ 0,1 % duty cycle | The whole band | | 40 |
| H | 433,050 MHz to 434,790 MHz | 10 mW | 10 % | The whole band | | 44b, 45b |
| I | 433,050 MHz to 434,790 MHz | 1 mW e.r.p. -13 dBm/10 kHz power spectral density for bandwidth modulation larger than 250 kHz | No requirement | The whole band | Audio and video applications are excluded. | 44a, 45a |
| J | 434,040 MHz to 434,790 MHz | 10 mW | No requirement | 25 kHz | Audio and video applications are excluded. | 45c |
| K | 863 MHz to 865 MHz | 25 mW e.r.p. | ≤ 0,1 % duty cycle or polite spectrum access | The whole band except for audio & video applications limited to 300 kHz | | 46a |

| | | | | | | |
|---|---|---|---|---|---|---|
| L | 865 MHz to 868 MHz | 25 mW e.r.p. | ≤ 1 % duty cycle or polite spectrum access | The whole band | | 47 |
| M | 868,000 MHz to 868,600 MHz | 25 mW e.r.p. | ≤ 1 % duty cycle or polite spectrum access | The whole band | | 48 |
| N | 868,700 MHz to 869,200 MHz | 25 mW e.r.p. | ≤ 0,1 % duty cycle or polite spectrum access | The whole sub-band | | 50 |
| P | 869,400 MHz to 869,650 MHz | 500 mW e.r.p. | ≤ 10 % duty cycle or polite spectrum access | The whole band | | 54 |
| P | 869,700 MHz to 870,000 MHz | 5 mW e.r.p. | No requirement | The whole band | Audio and video applications are excluded. | 56a |
| Q | 869,700 MHz to 870,000 MHz | 25 mW e.r.p. | ≤ 1 % duty cycle or polite spectrum access | The whole band | Analogue audio applications are excluded. Analogue video applications are excluded. | 56b |

## 4.2. Information recorded

| Parameter | Value | Notes |
|---|---|---|
| Operational Frequency band or bands | 863 MHz to 865 MHz<br>868 MHz to 868.6 MHz | Declared by the standard |
| Nominal Operating Frequency or Frequencies | 868.1MHz, 868.3MHz, 868.5MHz, 864.1MHz, 864.3MHz,864.5MHz | Declared by the manufacturer |
| Operating Channel width(s) - OCW | 200kHz | Declared by the manufacturer |

The Operating Channel Width (OCW) of the EUT: 200kHz, and two frequency edges values:

for 868.5MHz,

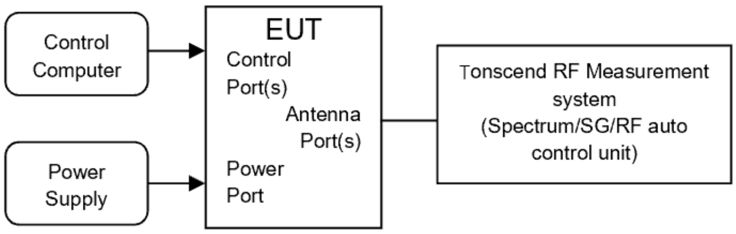
$F_{low}$=868.4 MHz and $F_{high}$=868.6 MHz,

for 864.1MHz

$F_{low}$=864 MHz and $F_{high}$=864.2 MHz, are declared by the manufacturer.

# 5. Unwanted Emissions in the Spurious Domain (Spectrum Mask)

## 5.1. Block diagram of test setup




## 5.2. Limits

Unwanted emissions for a TX mode:

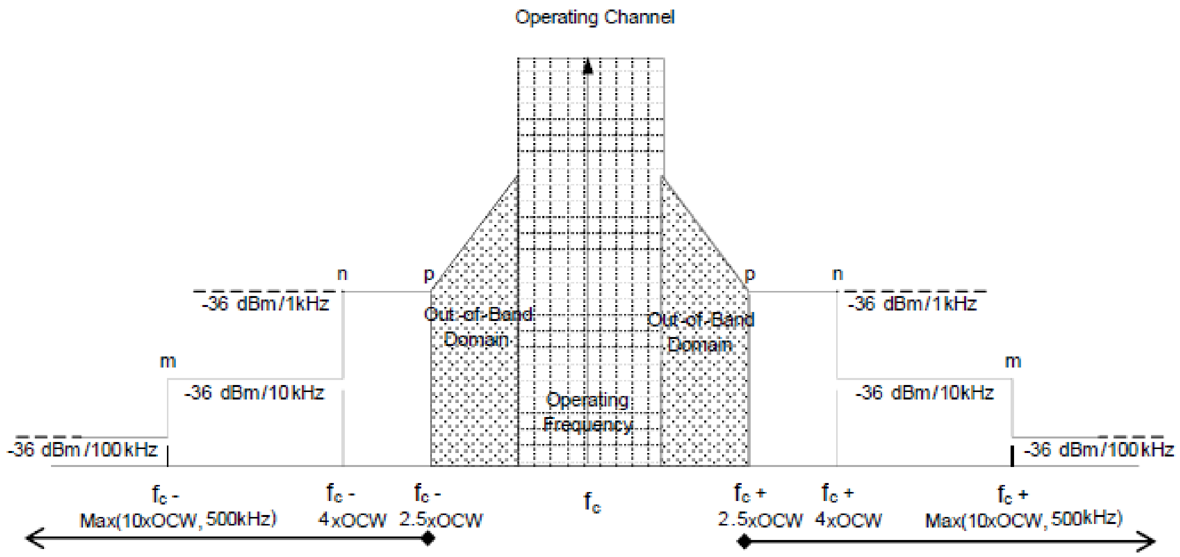



**Figure 7: Spectrum Mask for Unwanted Emissions in the Spurious Domain with reference BW**

## 5.3. Test procedure

1. Please refer to ETSI EN300220-1 Sub-clause 5.9.3.1/Sub-clause 5.9.3.2 for the test conditions.
2. Please refer to ETSI EN300220-1 Sub-clause 5.9.3.3/5.9.3.3.1 and Sub-clause 5.9.3.3/5.9.3.3.2 for the measurement method.
3. Parameters for TX Spurious Radiations Measurement:

| Operating Mode | Frequency Range | $RBW_{REF}$ (see note 2) |
|---|---|---|
| Transmit mode | 9 kHz ≤ f < 150 kHz | 1 kHz |
| | 150 kHz ≤ f < 30 MHz | 10 kHz |
| | 30 MHz ≤ f < $f_c$ - m | 100 kHz |
| | $f_c$ - m ≤ f < $f_c$ - n | 10 kHz |
| | $f_c$ - n ≤ f < $f_c$ - p | 1 kHz |
| | $f_c$ + p < f ≤ $f_c$ + n | 1 kHz |
| | $f_c$ + n < f ≤ $f_c$ + m | 10 kHz |
| | $f_c$ + m < f ≤ 1 GHz | 100 kHz |
| | 1 GHz < f ≤ 6 GHz | 1 MHz |

NOTE 1: f is the measurement frequency.
$f_c$ is the Operating Frequency.
m is 10 x OCW or 500 kHz, whichever is the greater.
n is 4 x OCW or 100 kHz, whichever is the greater.
p is 2,5 x OCW.
NOTE 2: If the value of RBW used for measurement is different from $RBW_{REF}$, use bandwidth correction from clause 4.3.10.1.

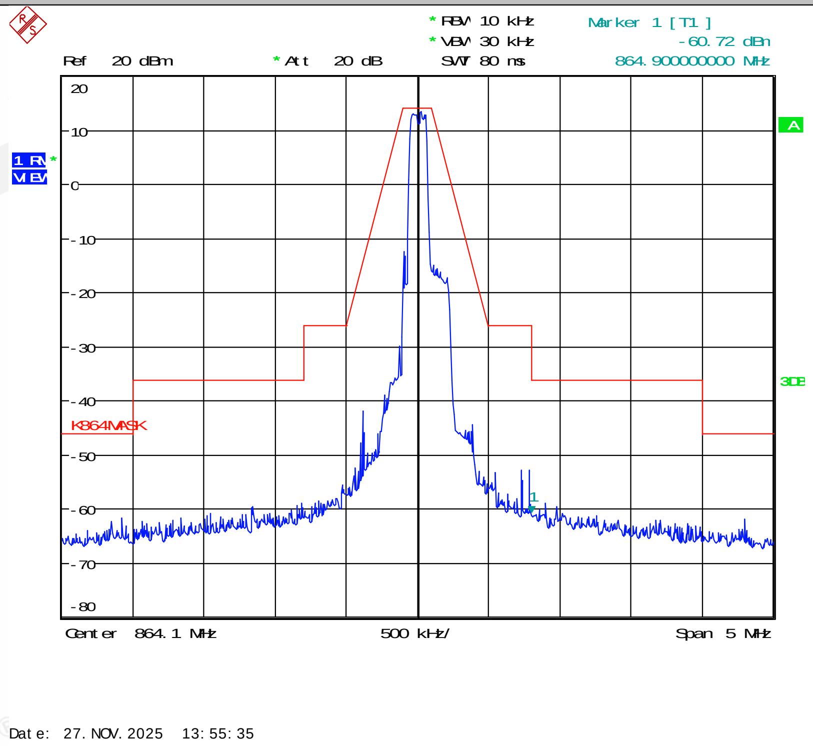
## 5.4. Test result

| | | | |
|---|---|---|---|
| Test Engineer: | Zeng Zhongyao | Test Site: | RF Measurement System 2# |
| Ambient Condition: | 23.5 ℃, 55%RH | Test Date: | 2025/11/27 - 2025/12/4 |
| Test Power Supply: | DC 3.3V from control board | Sample Number: | S25091517-003 |

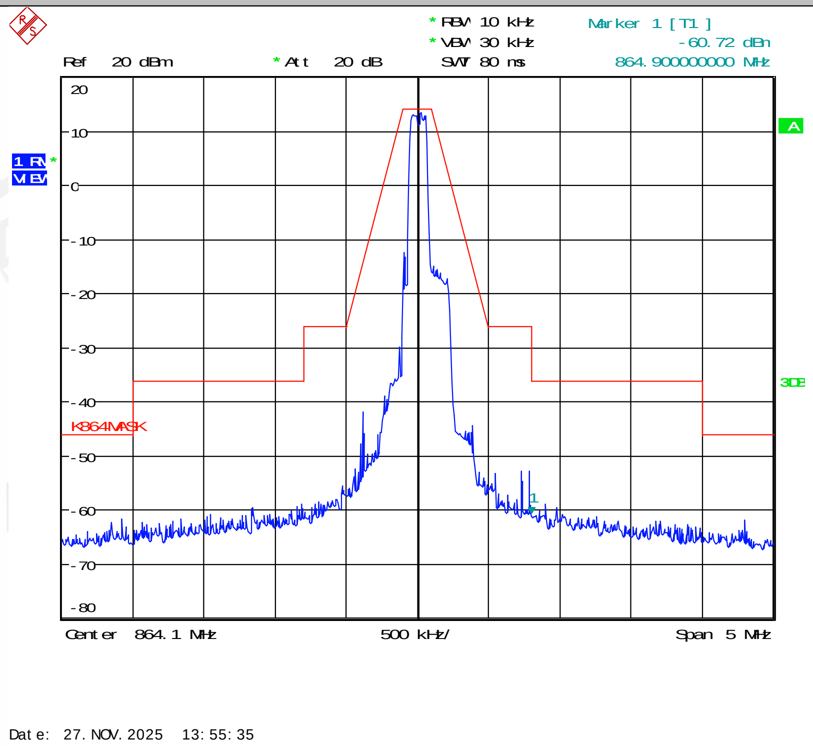
Unwanted emissions for a TX mode

| Test Conditions | Frequency Band [MHz] | Limit [dBm] | Verdict |
|---|---|---|---|
| NTNV | Low~fc-10*OCW | -36dB/100kHz | PASS |
| | fc-10*OCW~ fc-4*OCW | -36dB/10kHz | PASS |
| | fc-4*OCW~ fc-2.5*OCW | -36dB/1kHz | PASS |
| | fc-+2.5*OCW ~ fc-+4*OCW | -36dB/1kHz | PASS |
| | fc-+4*OCW~ fc-+10*OCW | -36dB/10kHz | PASS |
| | fc-+10*OCW~High | -36dB/100kHz | PASS |

## 5.5. Test graphs

| Spectrum Mask for Unwanted Emissions in the Spurious Domain_864.1MHz |
| --- |
|  |

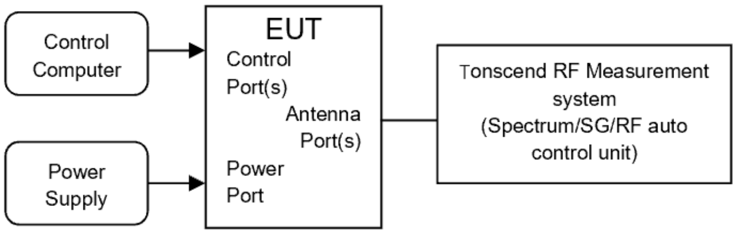


| Spectrum Mask for Unwanted Emissions in the Spurious Domain_868.5MHz |
| --- |
|  |

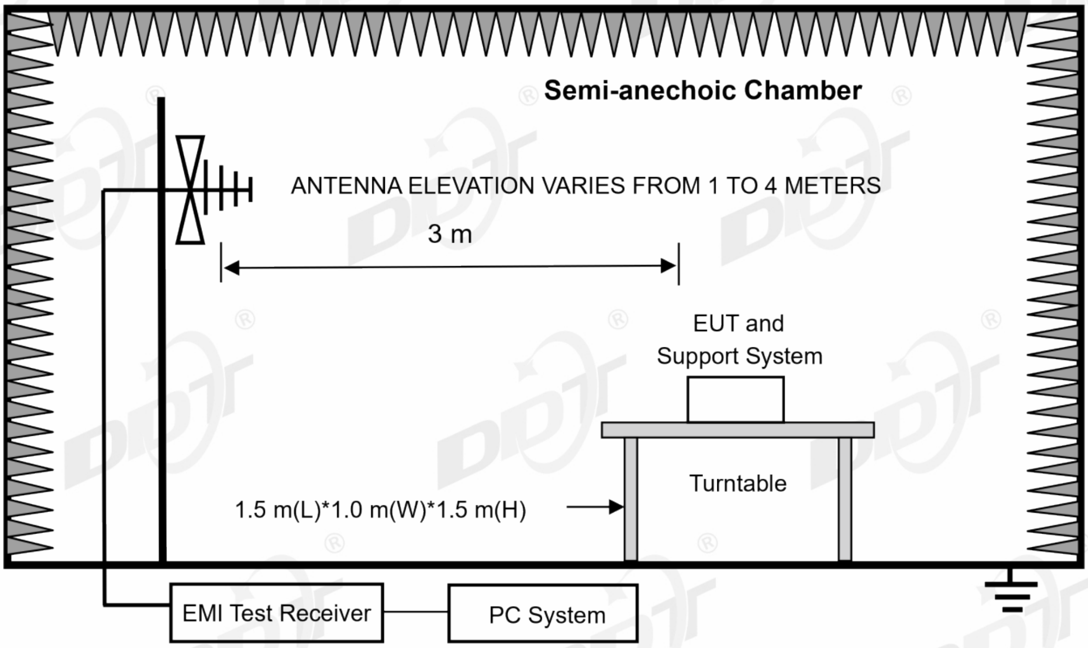
# 6. TX Effective Radiated Power

## 6.1. Block diagram of test setup

☐ Conducted Measurement Test Set-up




☒ Radiated Measurement Test Set-up



## 6.2. Limits

**Table B.1: EU wide harmonised national radio interfaces from 25 MHz to 1 000 MHz**

| Operational Frequency Band | | Maximum effective radiated power, e.r.p. | Channel access and occupation rules(e.g. Duty cycle or LBT + AFA) | Maximum occupied band width | Other usage restrictions | Band number from EC Decision 2017/1483/EU [2] |
|---|---|---|---|---|---|---|
| A | 26,957 MHz to 27,283 MHz | 10 mW e.r.p. | No requirement | The whole band | | 28 |
| B | 26,995 MHz, 27,045 MHz, 27,095 MHz, 27,145 MHz, 27,195 MHz | 100 mW e.r.p. | ≤ 0,1 % duty cycle | 10 kHz | Model control devices may operate without duty cycle restrictions. | 29, 30, 31, 32, 33 |
| C | 40,660 MHZ to 40,700 MHz | 10 mW e.r.p. | No requirement | The whole band | Video applications excluded. | 35 |
| D | 169,400 MHz to 169,475 MHz | 500 mW e.r.p. | ≤ 1,0 % duty cycle For metering devices duty cycle limit is 10 % | 50 kHz | | 37c |
| E | 169,4000 MHz to 169,4875 MHz | 10 mW | ≤ 0,1 % duty | The whole band | | 38 |
| F | 169,4875 MHZ to 169,5875 MHz | 10 mW | ≤ 0,001 % duty cycle Between 00.00 and 06.00 local time a duty cycle limit of 0,1 % may be used | The whole band | Equipment that concentrates or multiplexes individual equipment is excluded. | 39b |
| G | 169,5875 MHz to 169,8125 MHz | 10 mW | ≤ 0,1 % duty cycle | The whole band | | 40 |
| H | 433,050 MHz to 434,790 MHz | 10 mW | 10 % | The whole band | | 44b, 45b |
| I | 433,050 MHz to 434,790 MHz | 1 mW e.r.p. -13 dBm/10 kHz power spectral density for bandwidth modulation larger than 250 kHz | No requirement | The whole band | Audio and video applications are excluded. | 44a, 45a |
| J | 434,040 MHz to 434,790 MHz | 10 mW | No requirement | 25 kHz | Audio and video applications are excluded. | 45c |

| | | | | | | |
|---|---|---|---|---|---|---|
| K | 863 MHz to 865 MHz | 25 mW e.r.p. | ≤ 0,1 % duty cycle or polite spectrum access | The whole band except for audio & video applications limited to 300 kHz | | 46a |
| L | 865 MHz to 868 MHz | 25 mW e.r.p. | ≤ 1 % duty cycle or polite spectrum access | The whole band | | 47 |
| M | 868,000 MHz to 868,600 MHz | 25 mW e.r.p. | ≤ 1 % duty cycle or polite spectrum access | The whole band | | 48 |
| N | 868,700 MHz to 869,200 MHz | 25 mW e.r.p. | ≤ 0,1 % duty cycle or polite spectrum access | The whole sub-band | | 50 |
| P | 869,400 MHz to 869,650 MHz | 500 mW e.r.p. | ≤ 10 % duty cycle or polite spectrum access | The whole band | | 54 |
| P | 869,700 MHz to 870,000 MHz | 5 mW e.r.p. | No requirement | The whole band | Audio and video applications are excluded. | 56a |
| Q | 869,700 MHz to 870,000 MHz | 25 mW e.r.p. | ≤ 1 % duty cycle or polite spectrum access | The whole band | Analogue audio applications are excluded. Analogue video applications are excluded. | 56b |

## 6.3. Test procedure

☐ Conducted measurement) (substitution method)

Refer to ETSI EN300220-1 Sub-clause 5.2.2.1.2. for the test conditions.

☒ Radiated measurement (substitution method)

Refer to ETSI EN 300 220-1 annex C and Sub-clause 5.2.2.2.2 for the test conditions.

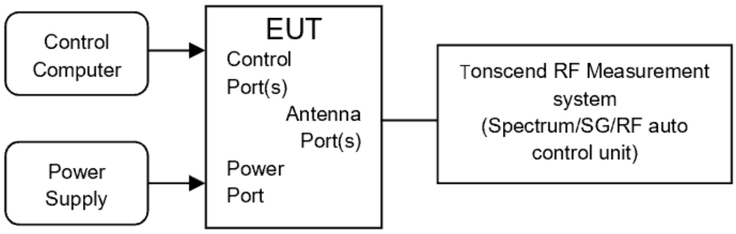
## 6.4. Test result

| | | | |
|---|---|---|---|
| Test Engineer: | Zeng Zhongyao | Test Site: | RF Measurement System 2# |
| Ambient Condition: | 23.5 ℃, 55%RH | Test Date: | 2025/11/27 - 2025/12/9 |
| Test Power Supply: | DC 3.3V from control board | Sample Number: | S25091517-003 |

| Operating Frequency [MHz] | Maximum effective radiated power | | Limit [mW] | Verdict |
|---|---|---|---|---|
| | dBm | mW | | |
| 864.1 | 13.37 | 21.73 | 25 | PASS |
| 868.5 | 13.75 | 23.71 | 25 | PASS |

# 7. Tx Maximum E.R.P. Power Spectral Density

## 7.1. Block diagram of test setup




## 7.2. Limits

The Maximum e.r.p. power spectral density shall not be greater than the value allowed in annex B or any NRI for the chosen operational frequency band(s).

## 7.3. Test procedure

(1) The test according to ETSI EN 300 220-1 V3.1.1 Clause 5.3.2.1.2.
(2) Connect EUT's antenna output to spectrum analyzer by RF cable, the path loss was compensated to the results.
(3) Set the spectrum analyzer as follows:

| | |
|---|---|
| Centre Frequency: | The centre frequency of the channel under test |
| Span: | Wide enough to cover the complete power envelope of the signal of the EUT (≥ Occupied Bandwidth). |
| Resolution BW: | 100 kHz (see note). |
| Video BW: | 100 kHz (see note). |
| Sweep time: | 1 minute. |
| Detector: | RMS |
| Trace Mode: | Max Hold. |
| Note: In case the regulatory parameter is expressed in dBm/10 kHz, RBW & VBW should be set to 10 kHz. | |

When the trace is complete, capture the trace, for example using the "View" option on the spectrum analyzer.
Find the peak value of the trace and place the analyzer marker on this peak. This level is recorded as the highest mean power (spectral power density) D in a 100 kHz band.
Alternatively, where a spectrum analyzer is equipped with a facility to measure spectral power density, this facility may be used to display the spectral power density D in dBm/100 kHz.

Where the spectrum analyzer bandwidth is non-Gaussian, a suitable correction factor shall be determined and applied.

The maximum e.r.p. spectral density is calculated from the above measured power density (D and the applicable antenna assembly gain "G" in dB relative to an ideal half wave dipole, according to below formula. If more than one antenna assembly is intended for this power setting, the gain of the antenna assembly with the highest gain shall be used.

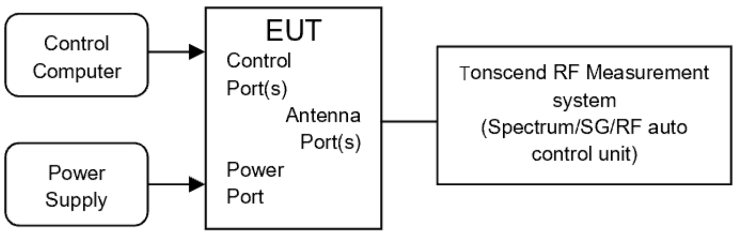
$$PD = D + G$$

## 7.4. Test result

Not Applicable. Only applies to EUT work within operational frequency band I.

# 8. Duty Cycle

## 8.1. Block diagram of test setup




## 8.2. Limit

Please refer to ETSI EN300220-1 Annex B

## 8.3. Test procedure

(1) Connect EUT's antenna output to spectrum analyzer though sutiable attenuator.

(2) Configure EUT work in normal trasmit mode.

(3) Measure the total transmit ON time of each transmit once triggered with spectrum analyzer set as below:

RBW= 10MHz; VBW=10MHz; Span: 0MHz; Detector: PK; Center frequency: transmit frequency.

(4) Duty cycle $=T_{on}/T_{on}+T_{off}$

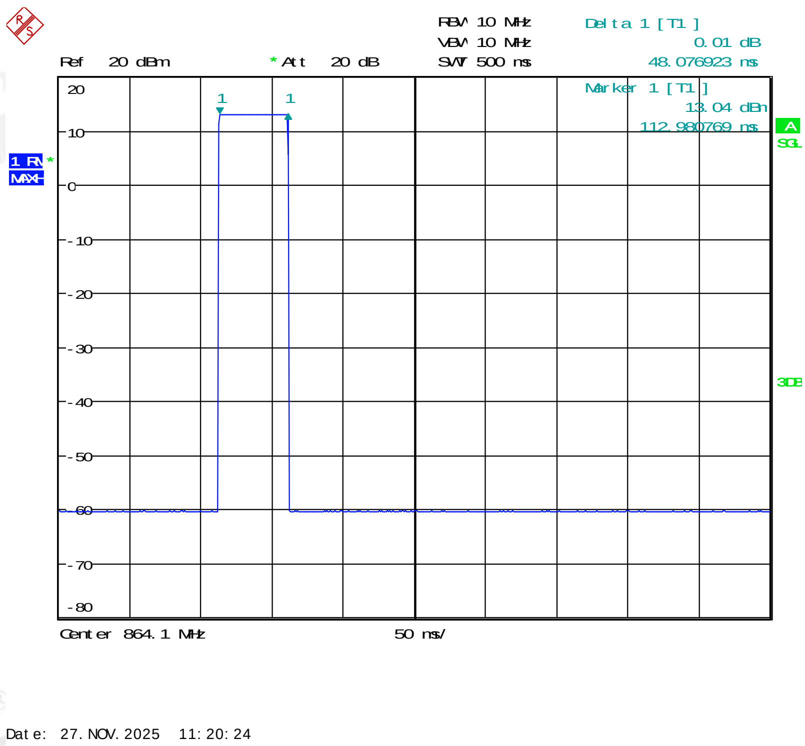
## 8.4. Test result

| | | | |
|---|---|---|---|
| Test Engineer: | Zeng Zhongyao | Test Site: | RF Measurement System 2# |
| Ambient Condition: | 23.5 ℃, 55%RH | Test Date: | 2025/11/27 - 2025/12/4 |
| Test Power Supply: | DC 3.3V from control board | Sample Number: | S25091517-003 |

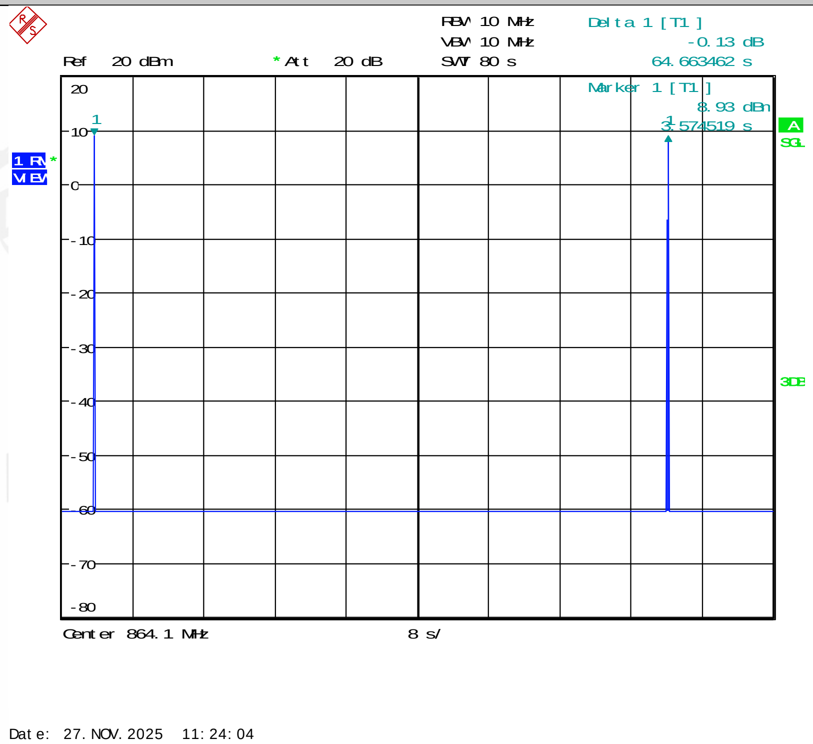
| Centre Frequency [MHz] | On time [ms] | Total time [ms] | Duty cycle [%] | limit | Verdict |
|---|---|---|---|---|---|
| 864.1 | 48.077 | 64663.462 | 0.07 | 0.1% | Pass |
| 868.5 | 48.077 | 63531.250 | 0.07 | 1% | Pass |

## 8.5. Test graphs

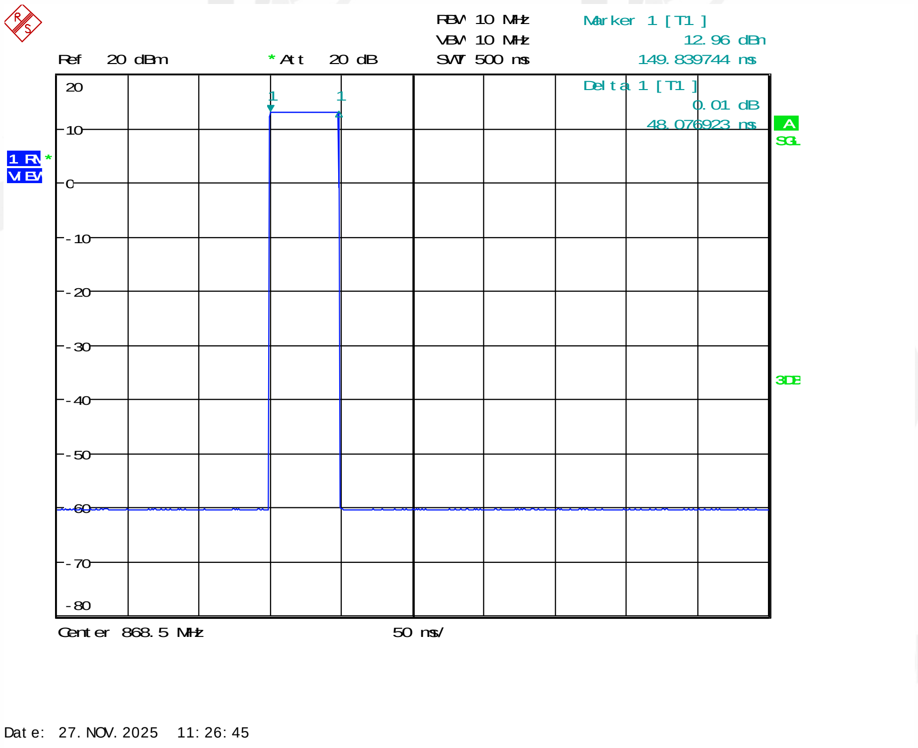
Duty cycle_ On time_864.1MHz



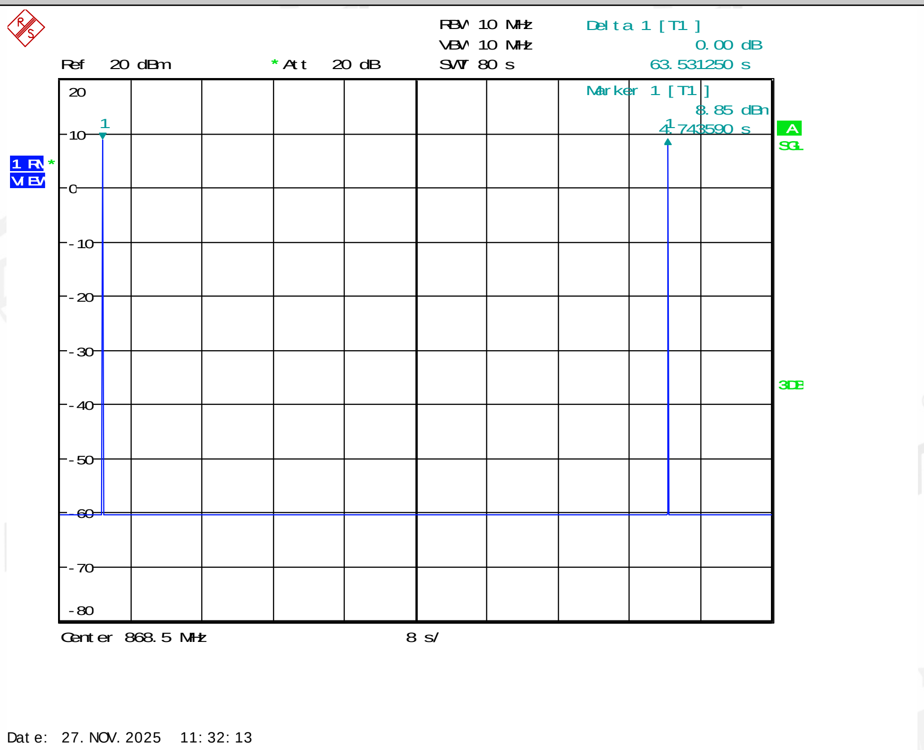

Duty cycle_ Total time_864.1MHz



Duty cycle_ On time_868.5MHz

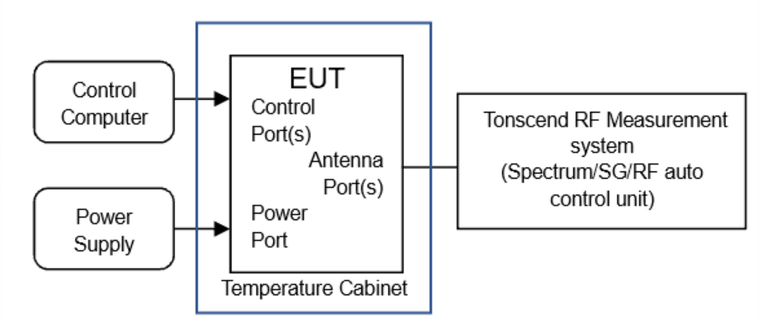



Duty cycle_ Total time_868.5MHz



# 9. TX Occupied Bandwidth

## 9.1. Block diagram of test setup




## 9.2. Limit

Limits apply under normal and extreme conditions

The Operating Channel shall be declared and shall reside entirely within the Operational Frequency Band. The Maximum Occupied Bandwidth at 99% shall reside entirely within the Operating Channel defined by $F_{low}$ and $F_{high}$.

Refer to clause 4.1, declared by the manufacturer.

## 9.3. Test Procedure

(1) Refer to ETSI EN300220-1 Sub-clause 5.6.3.1 for the test conditions.

(2) Refer to ETSI EN300220-1 Sub-clause 5.6.3.4 for the test Measurement procedure (Conducted measurement).

(3) The spectrum analyzer shall be configured as appropriate for the parameters shown in Table 12.

| Setting | Value | Notes |
|---|---|---|
| Centre frequency | The nominal Operating Frequency | The highest or lowest Operating Frequency as declared by the manufacturer |
| RBW | 1 % to 3 % of OCW without being below 100 Hz | |
| VBW | 3 x RBW | Nearest available analyser setting to 3 x RBW |
| Span | At least 2 x Operating Channel width | Span should be large enough to include all major components of the signal and its side bands |
| Detector Mode | RMS | |
| Trace | Max hold | |

## 9.4. Test Result

| | | | |
|---|---|---|---|
| Test Engineer: | Zeng Zhongyao | Test Site: | RF Measurement System 2# |
| Ambient Condition: | 23.5 ℃, 55%RH | Test Date: | 2025/11/27 - 2025/12/4 |
| Test Power Supply: | DC 3.3V from control board | Sample Number: | S25091517-003 |

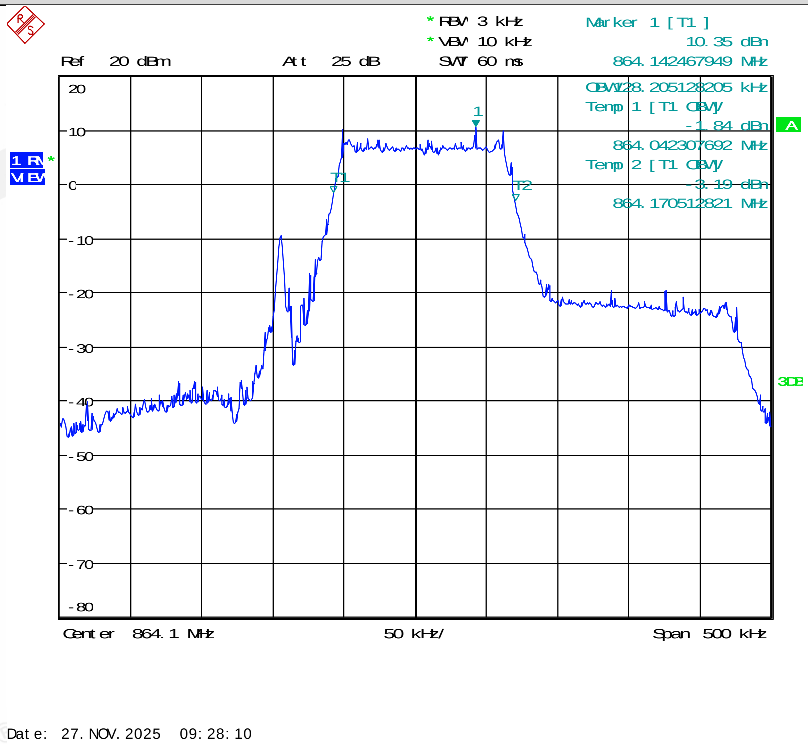
| Test Conditions (in Normal & Extreme) | | Modulation | $F_L$ (MHz) | $F_H$ (MHz) | Occupied Bandwidth (KHz) | Limit (MHz) | Result |
|---|---|---|---|---|---|---|---|
| Volt. | Temp. | | | | | | |
| Nor. Volt. | Nor. Temp. | CSS | 864.042 | 864.171 | 128.205 | 864-864.2 | PASS |
| Low Volt. | High Temp. | CSS | 864.042 | 864.171 | 128.205 | 864-864.2 | PASS |
| High Volt. | Low Temp. | CSS | 864.042 | 864.171 | 128.205 | 864-864.2 | PASS |
| High Volt. | High Temp. | CSS | 864.041 | 864.171 | 129.000 | 864-864.2 | PASS |
| Low Volt. | Low Temp. | CSS | 864.042 | 864.171 | 128.205 | 864-864.2 | PASS |
| Maximum Occupied Bandwidth=0.13 MHz | | | | | | | |

| Test Conditions (in Normal & Extreme) | | Modulation | $F_L$ (MHz) | $F_H$ (MHz) | Occupied Bandwidth (KHz) | Limit (MHz) | Result |
|---|---|---|---|---|---|---|---|
| Volt. | Temp. | | | | | | |
| Nor. Volt. | Nor. Temp. | CSS | 868.443 | 868.570 | 125.801 | 868.4-868.6 | PASS |
| Low Volt. | High Temp. | CSS | 868.443 | 868.570 | 126.603 | 868.4-868.6 | PASS |
| High Volt. | Low Temp. | CSS | 868.443 | 868.569 | 125.801 | 868.4-868.6 | PASS |
| High Volt. | High Temp. | CSS | 868.443 | 868.569 | 126.603 | 868.4-868.6 | PASS |
| Low Volt. | Low Temp. | CSS | 868.443 | 868.569 | 126.603 | 868.4-868.6 | PASS |
| Maximum Occupied Bandwidth=0.127 MHz | | | | | | | |

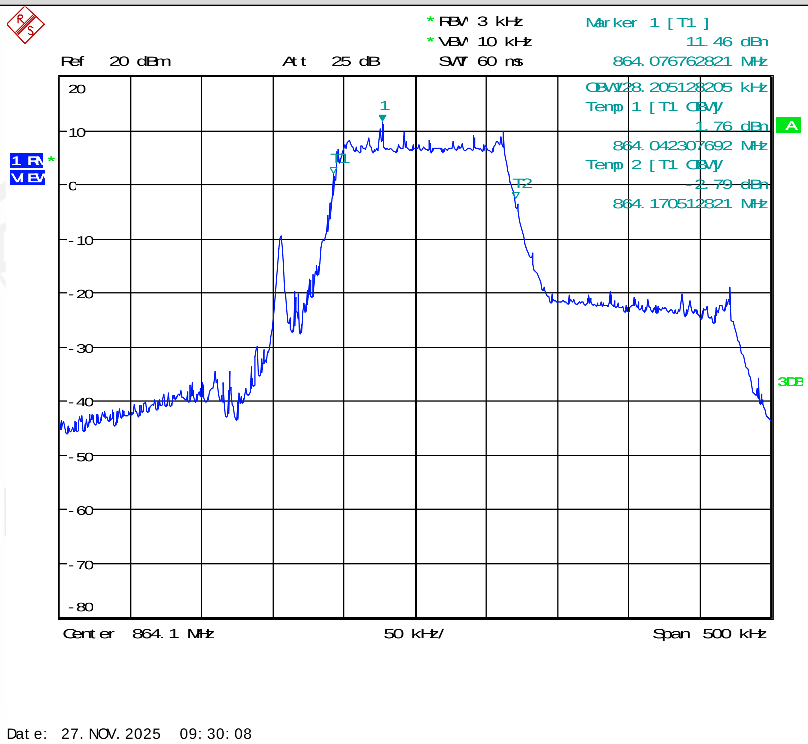
Note: The EUT is not capable of producing an unmodulated carrier.

## 9.5. Test graphs

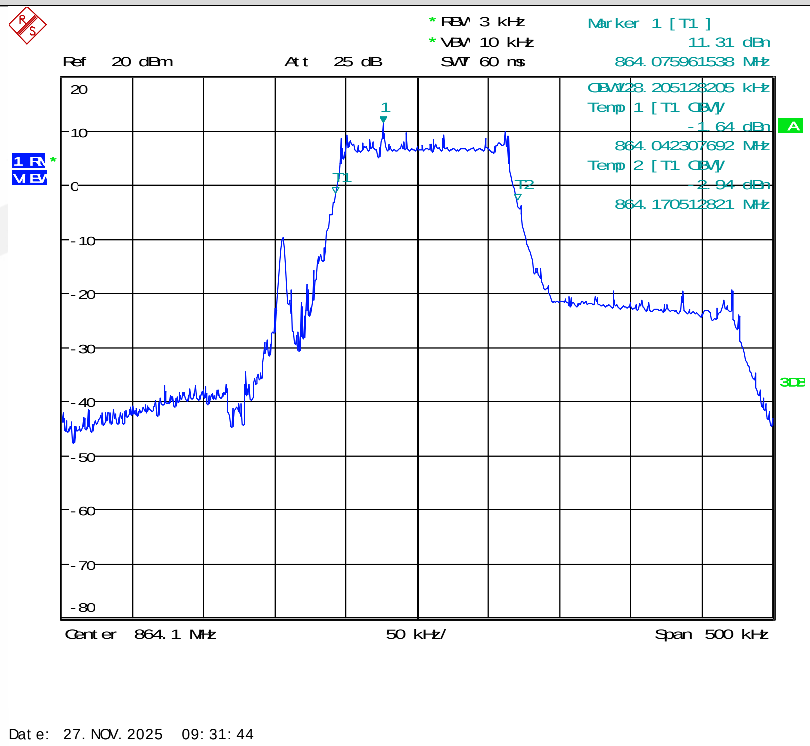
OBW_NTNV_864.1MHz



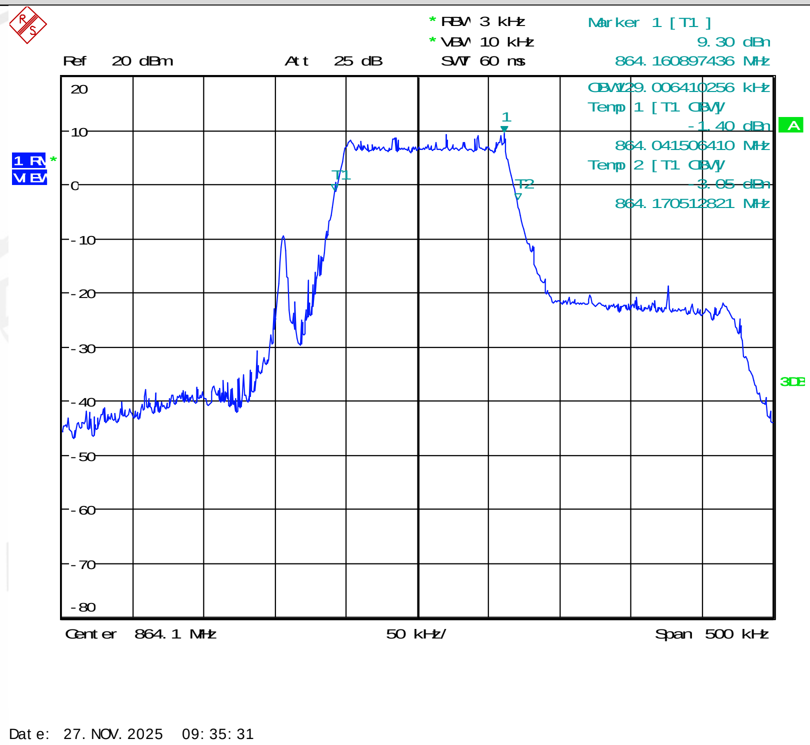

OBW_LTHV_864.1MHz



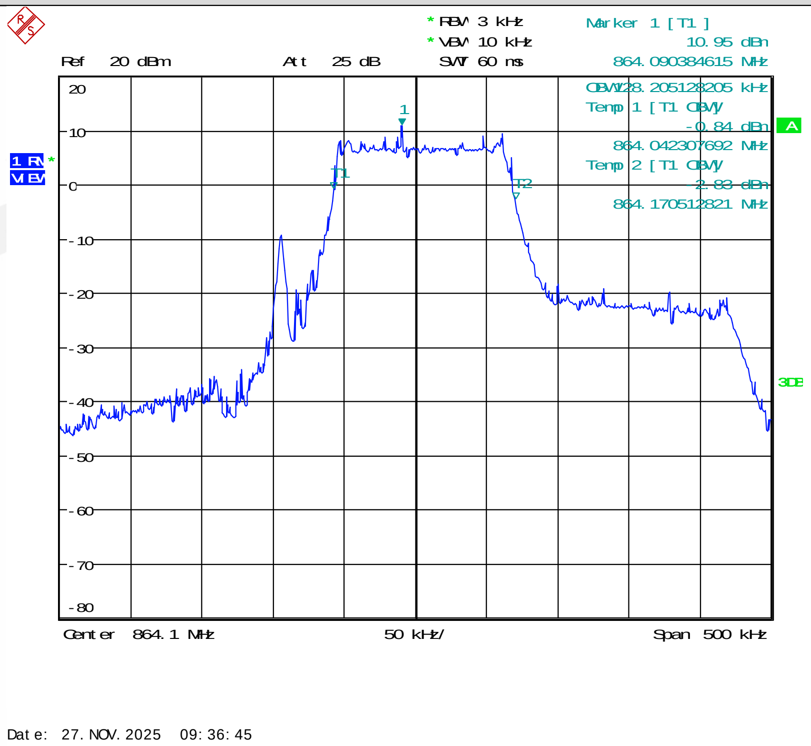
OBW_LTLV_864.1MHz



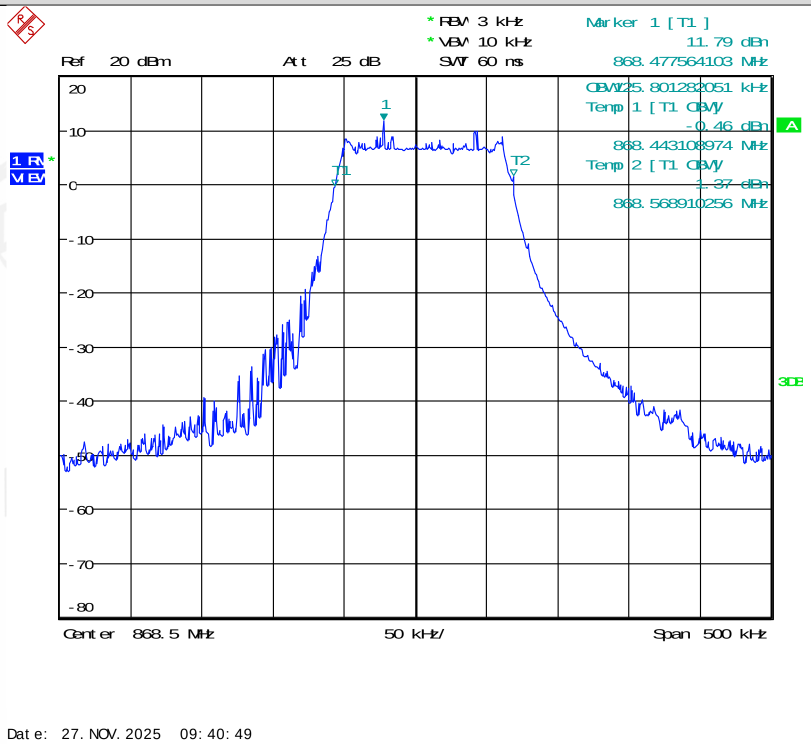

OBW_HTHV_864.1MHz



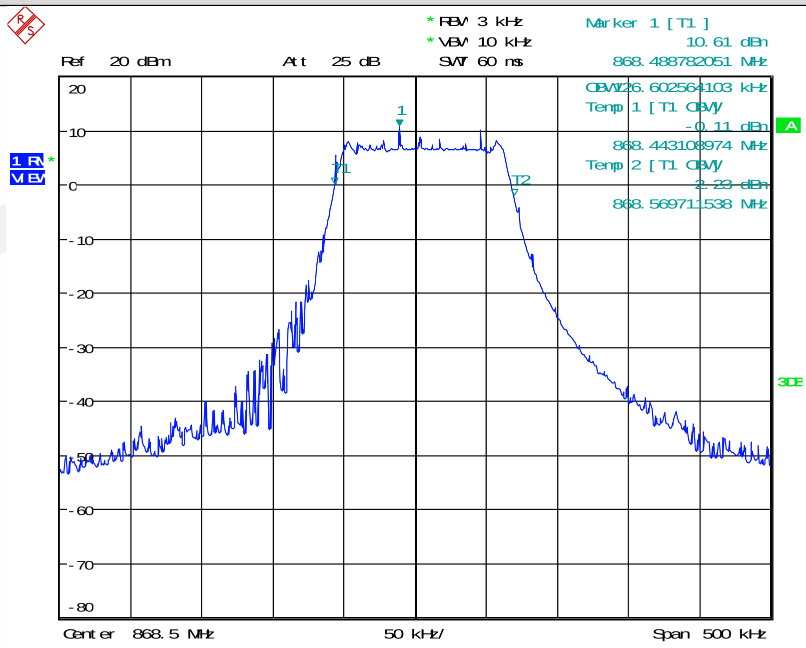
OBW_HTLV_864.1MHz




OBW_NTNV_868.5MHz

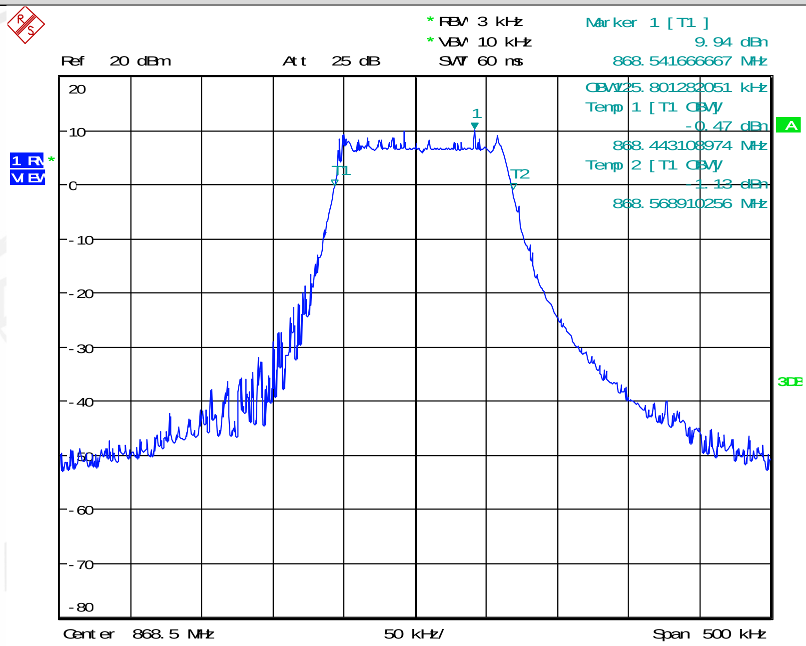


## OBW_LTHV_868.5MHz



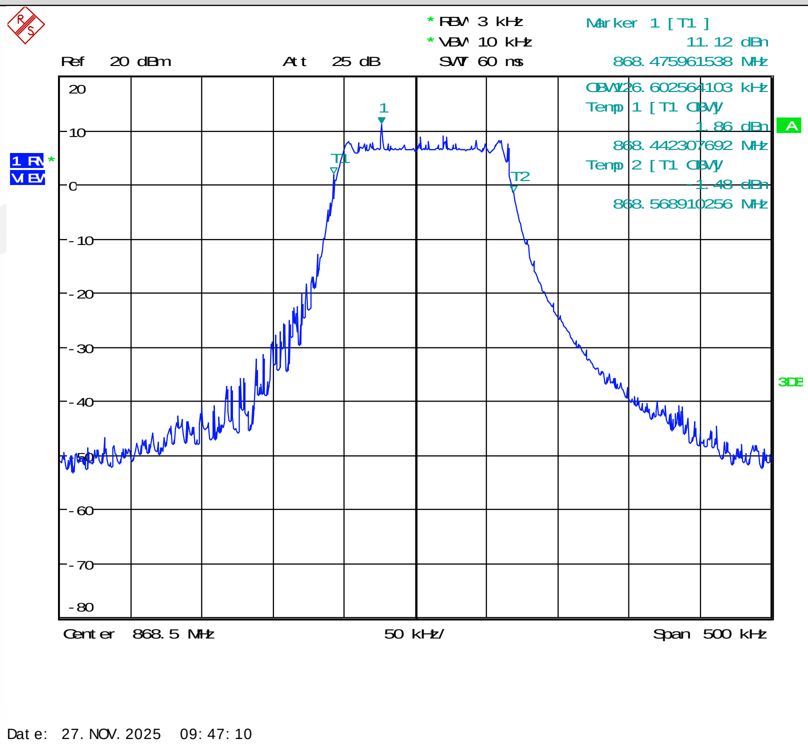

Date: 27.NOV.2025 09:43:57

## OBW_LTLV_868.5MHz

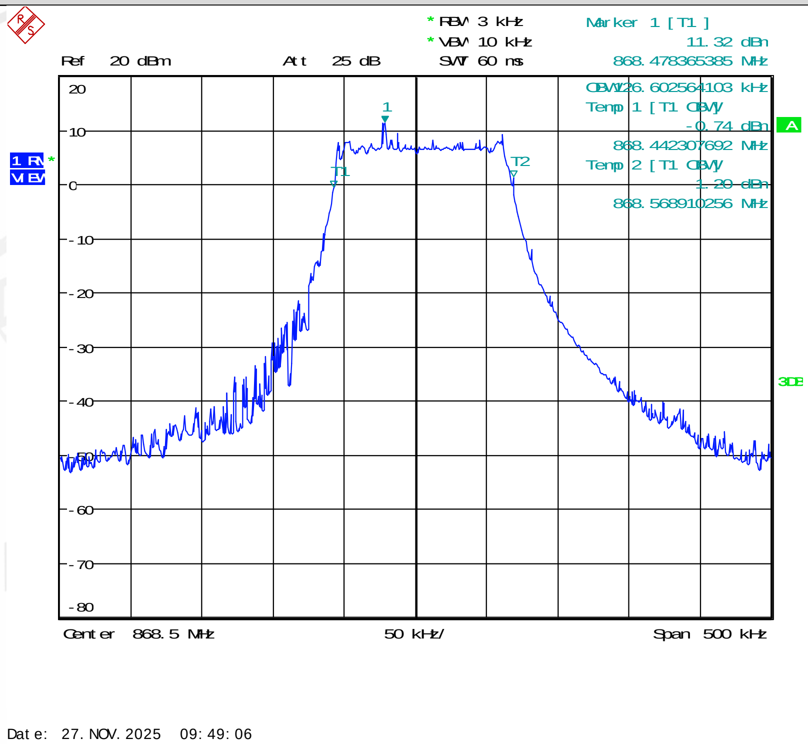



Date: 27.NOV.2025 09:45:52

OBW_HTHV_868.5MHz

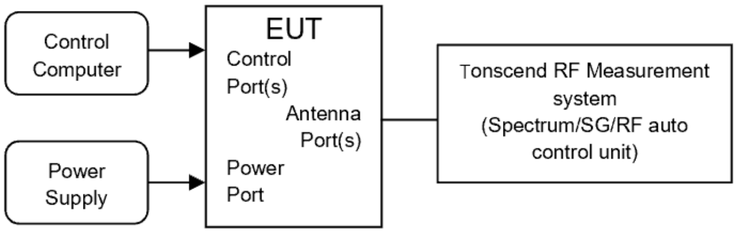



OBW_HTLV_868.5MHz



# 10. TX Out of Band Emissions

## 10.1. Block diagram of test setup




## 10.2. Limits

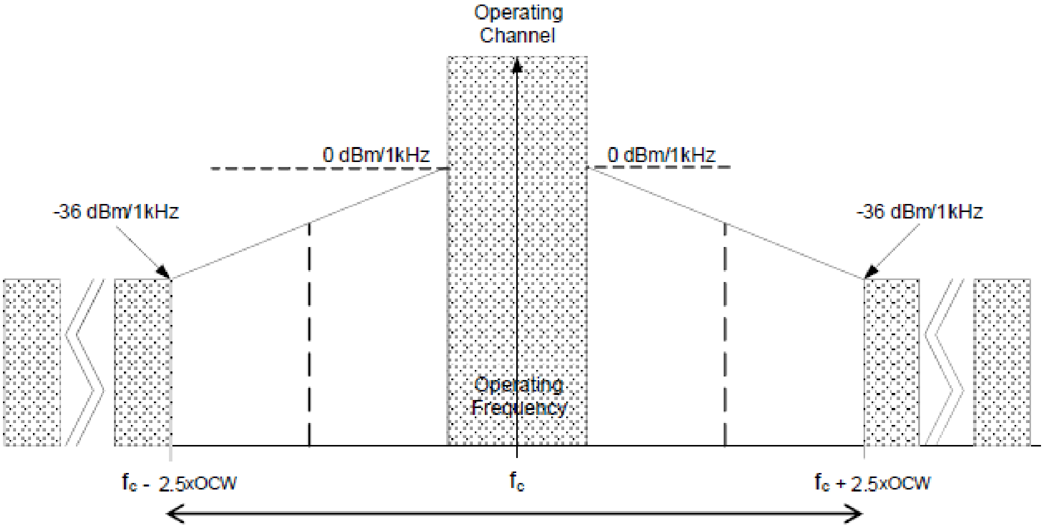



**Figure 5: Out Of Band Domain for Operating Channel with reference BW**

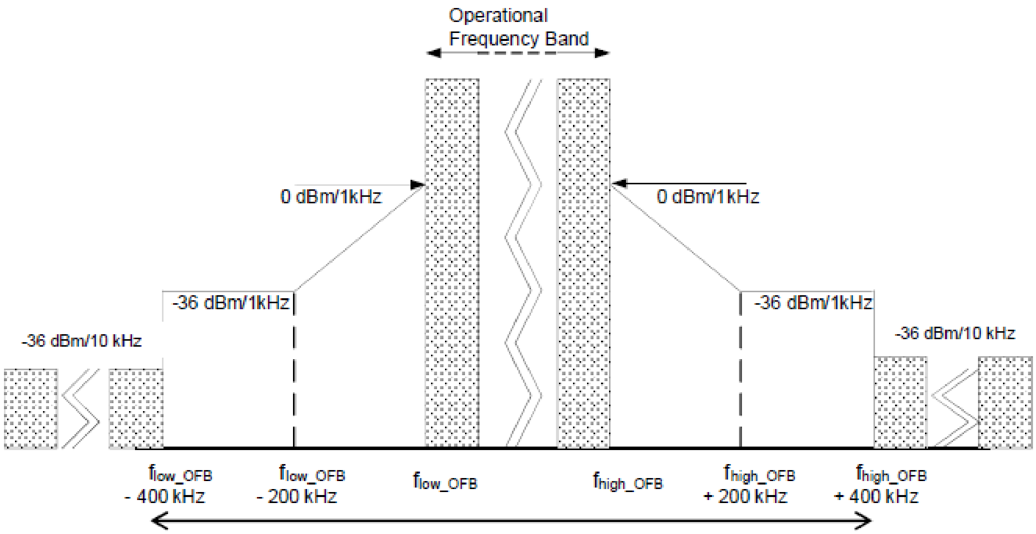



**Figure 6: Out Of Band Domain for Operational Frequency Band with reference BW**

## 10.3. Test procedure

Refer to ETSI EN300220-1 Sub-clause 5.8.3.2. for the test conditions.

Refer to ETSI EN300220-1 Sub-clause 5.8.3.4. for the test conditions.

RBW for Transient Measurement:

| Domain | Frequency Range | $RBW_{REF}$ | Max power limit |
|---|---|---|---|
| OOB limits applicable to Operational Frequency Band (See Figure 6) | $f \le f_{low\_OFB}$ - 400 kHz | 10 kHz | -36 dBm |
| | $F_{low\_OFB}$ - 400 kHz $\le f \le f_{low\_OFB}$ - 200 kHz | 1 kHz | -36 dBm |
| | $f_{low}$ - 200 kHz $\le f < f_{low\_OFB}$ | 1 kHz | See Figure 6 |
| | $f = f_{low\_OFB}$ | 1 kHz | 0 dBm |
| | $f = f_{high\_OFB}$ | 1 kHz | 0 dBm |
| | $F_{high\_OFB} < f \le f_{high\_OFB}$ + 200 kHz | 1 kHz | See Figure 6 |
| | $F_{high\_OFB}$ + 200 kHz $\le f \le f_{high\_OFB}$ + 400 kHz | 1 kHz | -36 dBm |
| | $F_{high\_OFB}$ + 400 kHz $\le f$ | 10 kHz | -36 dBm |
| OOB limits applicable to Operating Channel (See Figure 5) | $f = f_c$- 2.5 x OCW | 1 kHz | -36 dBm |
| | $f_c$ - 2,5 x OCW $\le f \le f_c$ - 0,5 x OCW | 1 kHz | See Figure 5 |
| | $f = f_c$ - 0,5 x OCW | 1 kHz | 0 dBm |
| | $f = f_c$ + 0,5 x OCW | 1 kHz | 0 dBm |
| | $f_c$ + 0,5 x OCW $\le f \le f_c$ + 2,5 x OCW | 1 kHz | See Figure 5 |
| | $f = f_c$+ 2,5 x OCW | 1 kHz | -36 dBm |

NOTE: f is the measurement frequency.
$f_c$ is the Operating Frequency.
$F_{low\_OFB}$ is the lower edge of the Operational Frequency Band.
$F_{high\_OFB}$ is the upper edge of the Operational Frequency Band.
OCW is the operating channel bandwidth.

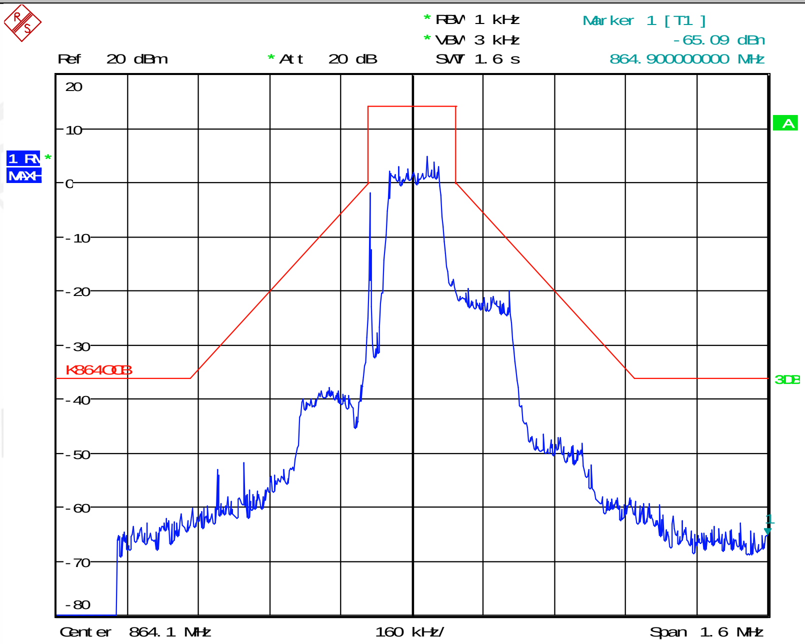
## 10.4. Test result

| | | | |
|---|---|---|---|
| Test Engineer: | Zeng Zhongyao | Test Site: | RF Measurement System 2# |
| Ambient Condition: | 23.5 ℃, 55%RH | Test Date: | 2025/11/27 - 2025/12/4 |
| Test Power Supply: | DC 3.3V from control board | Sample Number: | S25091517-003 |

| Centre Frequency [MHz] | Items | Verdict |
|---|---|---|
| 864.1 | Out Of Band Domain for Operating Channel with reference BW | PASS |
| | Out Of Band Domain for Operational Frequency Band with reference BW | PASS |
| 868.5 | Out Of Band Domain for Operating Channel with reference BW | PASS |
| | Out Of Band Domain for Operational Frequency Band with reference BW | PASS |

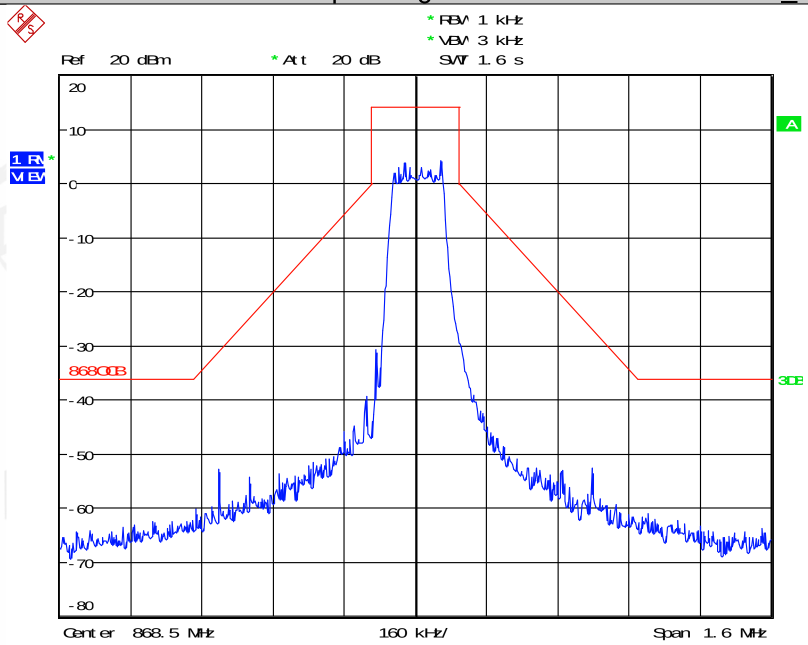
## 10.5. Test graphs

Out Of Band Domain for Operating Channel with reference BW_864.1MHz



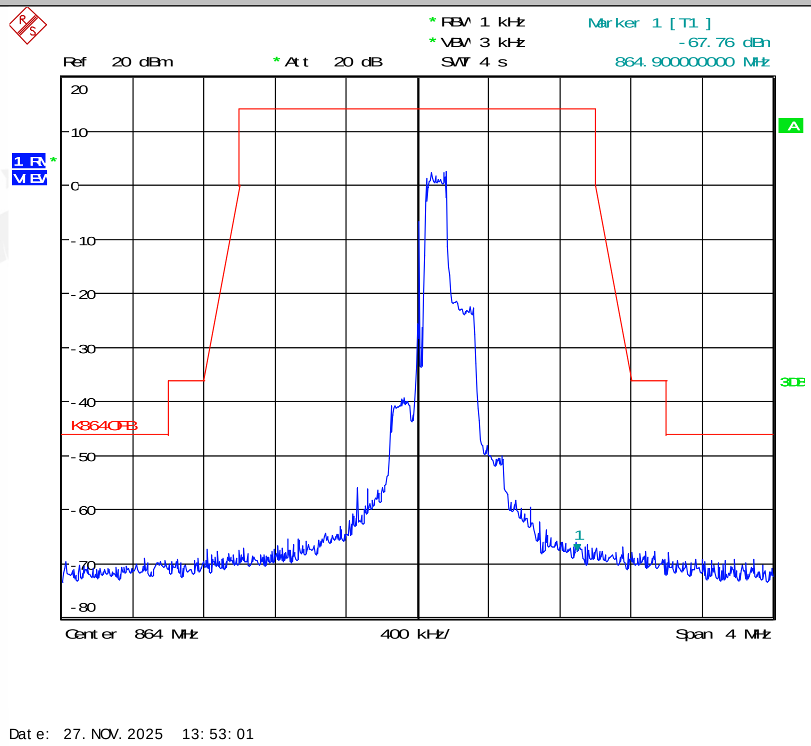

Date: 27.NOV.2025 13:46:04

Out Of Band Domain for Operating Channel with reference BW_868.5MHz

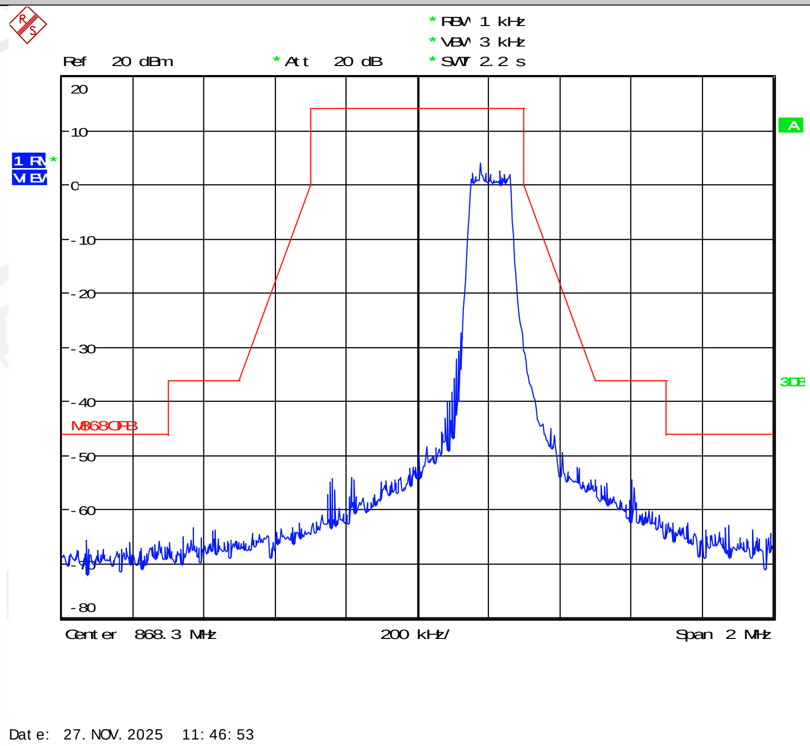



Date: 27.NOV.2025 11:41:03

Out Of Band Domain for Operational Frequency Band with reference BW_864.1MHz

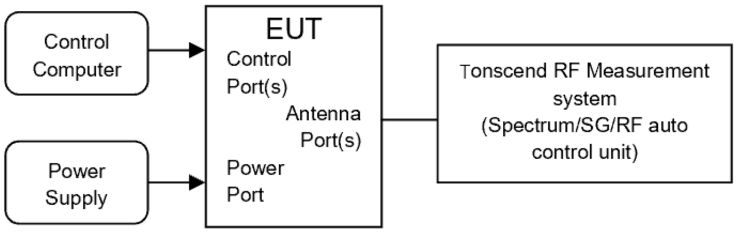



Out Of Band Domain for Operational Frequency Band with reference BW_868.5MHz



# 11. TX Transient

## 11.1. Block diagram of test setup




## 11.2. Limits

Transmitter transient power is power falling into frequencies other than the operating channel as a result of the transmitter being switched on and off, the transient power shall not exceed the values given in below table

| Absolute offset from Centre Frequency | $RBW_{REF}$ | Peak power limit applicable at measurement points |
|---|---|---|
| ≤ 400 kHz | 1 kHz | 0 dBm |
| > 400 kHz | 1 kHz | -27 dBm |

## 11.3. Test procedure

Refer to ETSI EN300220-1 Sub-clause 5.10.3. 1. for the test conditions.

Refer to ETSI EN300220-1 Sub-clause 5.10.3. 2. for the test conditions.

RBW for Transient Measurement:

| **Measurement points: offset from centre frequency** | **Analyser RBW** | **$RBW_{REF}$** |
|---|---|---|
| -0,5 x OCW - 3 kHz<br>0,5 x OCW + 3 kHz<br>Not applicable for OCW < 25 kHz | 1 kHz | 1kHz |
| ±12,5 kHz or ±OCW whichever is the greater | Max (RBW pattern 1, 3, 10 kHz) ≤ Offset frequency/6 (see note) | 1 kHz |
| -0,5 x OCW - 400 kHz<br>0,5 x OCW + 400 kHz | 100 kHz | 1 kHz |
| -0,5 x OCW -1 200 kHz<br>0,5 x OCW + 1 200 kHz | 300 kHz | 1 kHz |
| NOTE: Max (RBW pattern 1, 3, 10 kHz) means the maximum bandwidth that falls into the commonly implemented 1, 3, 10 kHz RBW filter bandwidth incremental pattern of spectrum analysers.<br>EXAMPLE: If OCW is 25 kHz then the RBW value corresponding to one OCW offset frequency is 3 kHz. The rest of the analyser settings are listed in Table 25, and if OCW is 250 kHz then the RBW value corresponding to one OCW offset frequency is 30 kHz. | | |

Parameters for Transient Measurement:

| Spectrum Analyser Setting | Value | Notes |
|---|---|---|
| VBW/RBW | 10 | At higher RBW values VBW may be clipped to its maximum value |
| Sweep time | 500 ms | |
| RBW filter | Gaussian | |
| Trace Detector Function | RMS | |
| Trace Mode | Max hold | |
| Sweep points | 501 | |
| Measurement mode | Continuous sweep | |
| NOTE: The ratio between the number of sweep points and the sweep time shall be the same ratio as above if different number of sweep points is used. | | |

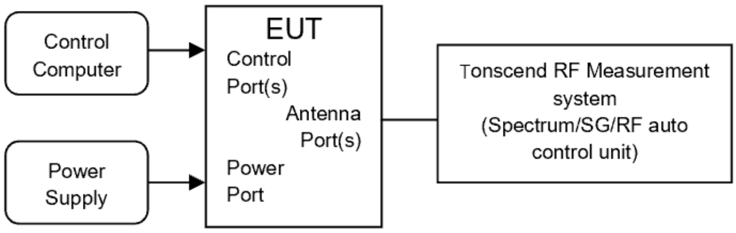
## 11.4. Test result

| | | | |
|---|---|---|---|
| Test Engineer: | Zeng Zhongyao | Test Site: | RF Measurement System 2# |
| Ambient Condition: | 23.5 ℃, 55%RH | Test Date: | 2025/11/27 - 2025/12/4 |
| Test Power Supply: | DC 3.3V from control board | Sample Number: | S25091517-003 |

| Offset from center frequency | Center frequency [MHz] | Value [dBm] | Limit [dBm] | Verdict |
|---|---|---|---|---|
| -0.5 * OCW-3kHz | -34.87 | -34.87 | 0 | PASS |
| 0.5 * OCW+3kHz | -27.93 | -27.93 | 0 | PASS |
| -OCW | -42.23 | -42.23 | 0 | PASS |
| +OCW | -27.28 | -27.28 | 0 | PASS |
| -0.5 * OCW-400kHz | -45.55 | -45.55 | -27 | PASS |
| 0.5 * OCW+400kHz | -47.66 | -47.66 | -27 | PASS |
| -0.5 * OCW-1200kHz | -41.57 | -41.57 | -27 | PASS |
| 0.5 * OCW+1200kHz | -41.07 | -41.07 | -27 | PASS |

# 12. TX Adjacent Channel Power

## 12.1. Block diagram of test setup




## 12.2. Limits

The adjacent channel power shall not exceed the maximum values give below:

| / | Test conditions | Adjacent Channel power integrated power 0.7*OCW | Alternate adjacent channel power integrated over 0.7*OCW |
|---|---|---|---|
| OCW<20 kHz | Normal | -20 dBm | -20 dBm |
| | Extreme | -15 dBm | -20 dBm |
| OCW≥20 kHz | Normal | -37 dBm | -40 dBm |
| | Extreme | -32 dBm | -37 dBm |

## 12.3. Test procedure

Pls. refer to ETSI EN300220-1 Sub-clause 5.11.3. 1. for the test conditions.

Pls. refer to ETSI EN300220-1 Sub-clause 5.11.3. 3/ 5.11.3.4 for the test conditions

## 12.4. Test result

Not Applicable. Only applies to EUT with OCW ≤ 25 kHz.

# 13. TX Behaviour Under Low Voltage Conditions

## 13.1. Limits

The equipment shall either:

(1) Remain in the Operating Channel without exceeding any applicable(e.g. duty cycle);or

(2) Reduce its effective radiated power below the spurious emission limits without exceeding any applicable limits (e.g. Duty Cycle) or

(3) Shut down, (e.g. no emission above EMC levels) as the voltage falls

(4) below the providers declared operating voltage

## 13.2. Test procedure

Refer to ETSI EN300220-1 Sub-clause 5.12.3. 1. for the test conditions.

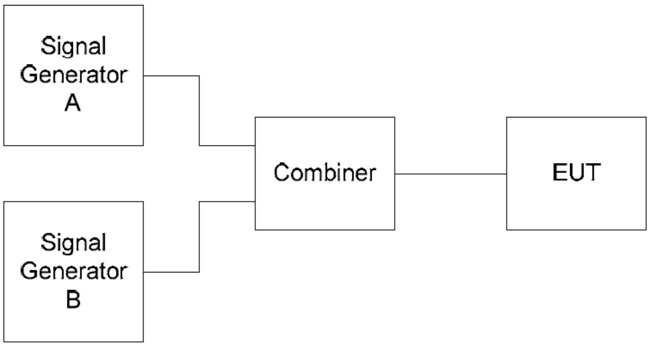
Refer to ETSI EN300220-1 Sub-clause 5.12.3. 2. for the test conditions.

## 13.3. Test result

Not Applicable. Only applies to battery powered EUT.

# 14. RX Blocking

## 14.1. Block diagram of test setup




## 14.2. Limits

Blocking is a measure of the capability of the receiver to receive a wanted modulated signal without exceeding a given degradation due to the presence of an unwanted input signal at any frequencies other than those of the spurious responses or the adjacent channels or bands.

The blocking level shall not be less than the values given in below table, except at frequencies on which spurious responses are found:

**Table 40: Blocking level parameters for RX category 3**

| Requirement | Limits |
|---|---|
| | Receiver category 3 |
| Blocking at ±2 MHz from OC edge $f_{high}$ and $f_{low}$ | ≥ -80 dBm |
| Blocking at ±10 MHz from OC edge $f_{high}$ and $f_{low}$ | ≥ -60 dBm |
| Blocking at ±5 % of Centre Frequency or 15 MHz, whichever is the greater | ≥ -60 dBm |

**Table 41: Blocking level parameters for RX category 2**

| Requirement | Limits |
|---|---|
| | Receiver category 2 |
| Blocking at ±2 MHz from OC edge $f_{high}$ and $f_{low}$ | ≥ -69 dBm |
| Blocking at ±10 MHz from OC edge $f_{high}$ and $f_{low}$ | ≥ -44 dBm |
| Blocking at ±5 % of Centre Frequency or 15 MHz, whichever is the greater | ≥ -44 dBm |

**Table 42: Blocking level parameters for RX category 1.5**

| Requirement | Limits |
|---|---|
| | Receiver category 1.5 |
| Blocking at ±2 MHz from OC edge $f_{high}$ and $f_{low}$ | ≥ -43 dBm |
| Blocking at ±10 MHz from OC edge $f_{high}$ and $f_{low}$ | ≥ -33 dBm |
| Blocking at ±5 % of Centre Frequency or 15 MHz, whichever is the greater | ≥ -33 dBm |

**Table 43: Blocking level parameters for RX category 1**

| Requirement | Limits |
|---|---|
| | Receiver category 1 |
| Blocking at ±2 MHz from Centre Frequency | ≥ -20 dBm |
| Blocking at ±10 MHz from Centre Frequency | ≥ -20 dBm |
| Blocking at ±5 % of Centre Frequency or 15 MHz, whichever is the greater | ≥ -20 dBm |

## 14.3. Test procedure

(1) Configure EUT and test equipment as clause 13.2, EUT’s antenna output was connected to spectrum analyzer by RF cable. The RF cable’s loss was input to spectrum analyzer as amplitude offset.

(2) Follow the test method described in clause 5.18.6.3 of EN300 220-1 to measure out the blocking level of transmitter.

## 14.4. Test result

| Test Engineer: | Zeng Zhongyao | Test Site: | RF Measurement System 2# |
|---|---|---|---|
| Ambient Condition: | 23.5 ℃, 55%RH | Test Date: | 2025/11/27 - 2025/12/4 |
| Test Power Supply: | DC 3.3V from control board | Sample Number: | S25091517-003 |

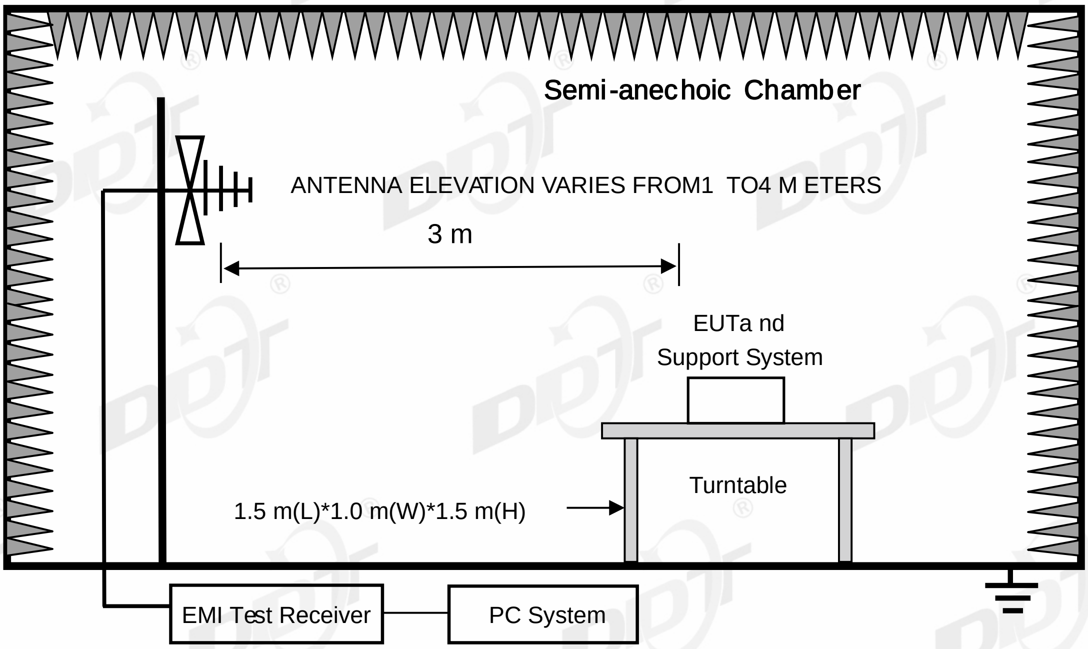
| Signal A | | Signal B | Blocking level (dBm) | Limit (dBm) | Result |
|---|---|---|---|---|---|
| Operation Frequency (MHz) | Test level（dB） | Blocking Frequency | | | |
| 864.1 | -99.6 | +2MHz | -30.1 | ≥-69 dBm | Pass |
| | | -2MHz | -29.3 | ≥-69 dBm | Pass |
| | | +10MHz | -22.4 | ≥-44 dBm | Pass |
| | | -10MHz | -23.9 | ≥-44 dBm | Pass |
| | | +5% | -20.8 | ≥-44 dBm | Pass |
| | | -5% | -20.6 | ≥-44 dBm | Pass |

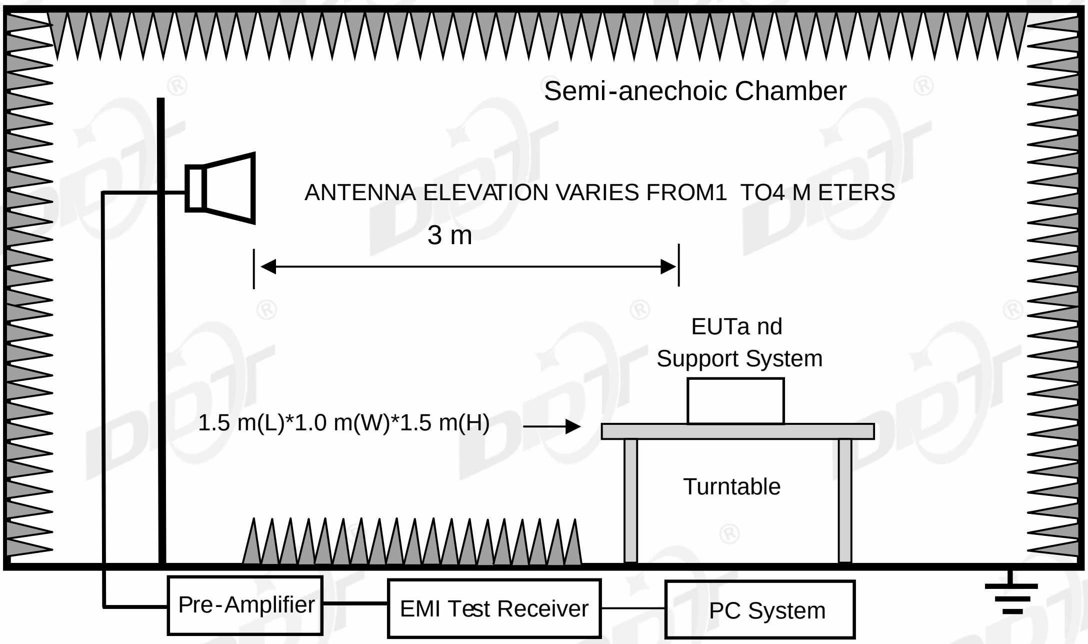
# 15. Unwanted emissions in the spurious domain

## 15.1. Test equipment

| Equipment | Manufacturer | Model No. | Serial No. | Cal Due To | Cal. Interval |
|---|---|---|---|---|---|
| High pass filter | Micro-Tronics | HPM50102 | DDT-ZC00561 | 2026/03/28 | 1 Year |
| Radiation disturbance fully automated test software | Tonscend | JS32-RE | DDT-ZC02739 | / | / |
| Pre-amplifier | SONOMA | 310N | DDT-ZC01969 | 2026/07/06 | 1 Year |
| RF Cable | N/A | W24.02 HL-562 | DDT-ZC04022 | 2026/03/28 | 1 Year |
| RF cable | Yuhu Technology | JCTB810-NJ-NJ-9M | DDT-ZC02538 | 2026/03/28 | 1 Year |
| RF cable | Zhongke Junchuang | JCT26S-NJ-NJ-1.5M | DDT-ZC02762 | / | / |
| RF cable | Yuhu Technology | ZT26S-SMAJ-SMAJ-1M | DDT-ZC02037 | 2026/10/10 | 1 Year |
| RF Cable | N/A | W13.02 AP1-X2 | DDT-ZC04023 | 2026/03/28 | 1 Year |
| Pre-amplifier | COM-POWER | PAM-840A | DDT-ZC01693 | 2026/03/28 | 1 Year |
| Pre-amplifier | COM-POWER | PAM-118A | DDT-ZC01293 | 2026/08/10 | 1 Year |
| EMI TEST RECEIVER | R&S | ESU26 | DDT-ZC01909 | 2026/03/28 | 1 Year |
| Micro-Tronics filters | REBES | BRM50702 | DDT-ZC03242 | / | / |
| High pass filter | Micro-Tronics | HPM50108 | DDT-ZC00560 | 2026/03/28 | 1 Year |
| High Pass filter | Xi'an Xingbo | XBLBQ-GTA67 | DDT-ZC02179 | 2026/03/28 | 1 Year |
| Micro-Tronics filters | REBES | BRM50716 | DDT-ZC03240 | / | / |
| Broad-Band Horn Antenna | Schwarzbeck | BBHA 9170 | DDT-ZC00506 | 2026/04/01 | 1 Year |
| Trilog Broadband Antenna | Schwarzbeck | VULB 9163 | DDT-ZC02050 | 2026/07/25 | 1 Year |
| Hochgewinn-Hornantenne | SCHWARZBECK | BBHA 9120 D | DDT-ZC02129 | 2026/08/11 | 1 Year |
| Active Loop Antenna | Schwarzbeck | FMZB1519 | DDT-ZC00524 | 2026/08/18 | 1 Year |
| PSA Series Spectrum Analyzer | Agilent | E4447A | DDT-ZC00517 | 2026/03/28 | 1 Year |

## 15.2. Block diagram of test setup



## 15.3. Limits

Unwanted emissions for all other modes:

| State | 47 MHz to 74 MHz<br>87.5 MHz to 118 MHz<br>174 MHz to 230 MHz<br>470 MHz to 862 MHz | Other frequencies<br>≤1 000 MHz | Frequencies<br>>1 000 MHz |
|---|---|---|---|
| TX mode | -54dBm | -36dBm | -30dBm |
| RX and all other modes | -57dBm | -57dBm | -47dBm |

## 15.4. Assistant equipment used for test

| Assistant equipment | Manufacturer | Model number | Description | other |
|---|---|---|---|---|
| Laptop | Lenovo | 00425-00000-00002-AA135 | NA | / |

## 15.5. Test procedure

1. Please refer to ETSI EN300220-1 Sub-clause 5.9.3.1/Sub-clause 5.9.3.2 for the test conditions.
2. Please refer to ETSI EN300220-1 Sub-clause 5.9.3.3/5.9.3.3.1 and Sub-clause 5.9.3.3/5.9.3.3.2 for the measurement method.
3. Parameters for TX Spurious Radiations Measurement:

| Operating Mode | Frequency Range | $RBW_{REF}$ (see note 2) |
|---|---|---|
| Transmit mode | 9 kHz ≤ f < 150 kHz | 1 kHz |
| | 150 kHz ≤ f < 30 MHz | 10 kHz |
| | 30 MHz ≤ f < $f_c$ - m | 100 kHz |
| | $f_c$ - m ≤ f < $f_c$ - n | 10 kHz |
| | $f_c$ - n ≤ f < $f_c$ - p | 1 kHz |
| | $f_c$ + p < f ≤ $f_c$ + n | 1 kHz |
| | $f_c$ + n < f ≤ $f_c$ + m | 10 kHz |
| | $f_c$ + m < f ≤ 1 GHz | 100 kHz |
| | 1 GHz < f ≤ 6 GHz | 1 MHz |

NOTE 1: f is the measurement frequency.
$f_c$ is the Operating Frequency.
m is 10 x OCW or 500 kHz, whichever is the greater.
n is 4 x OCW or 100 kHz, whichever is the greater.
p is 2,5 x OCW.

NOTE 2: If the value of RBW used for measurement is different from $RBW_{REF}$, use bandwidth correction from clause 4.3.10.1.

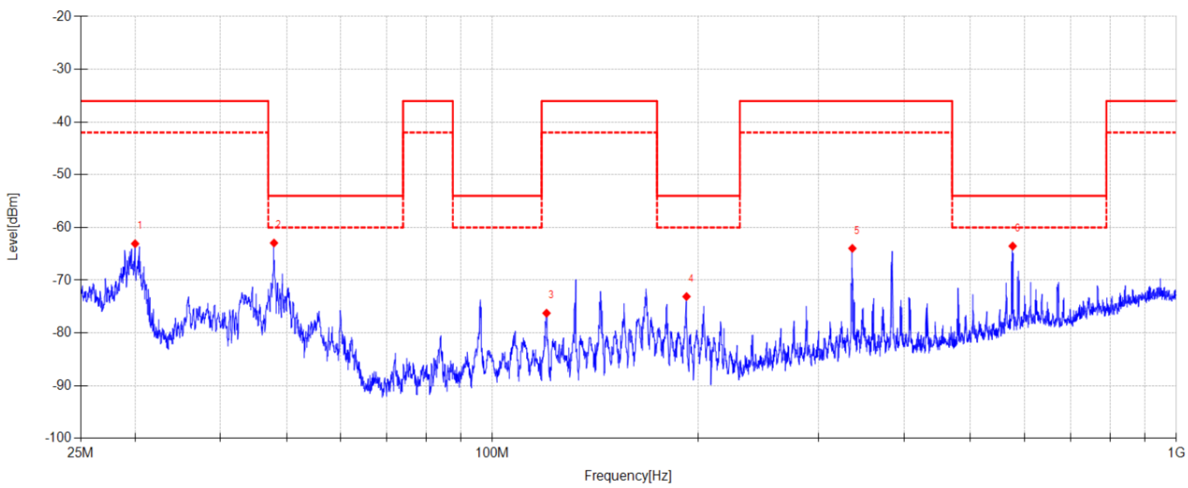
## 15.6. Test result

**PASS. (See below detailed test result)**

## 15.7. Test data

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-12-16 | **Tested By**: | Li Xiongbin |
| **Test Mode**: | TX 864.1MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Below 1G 868\33 | | |
| **Memo**: | Sample Number:S25091517-003 | | |

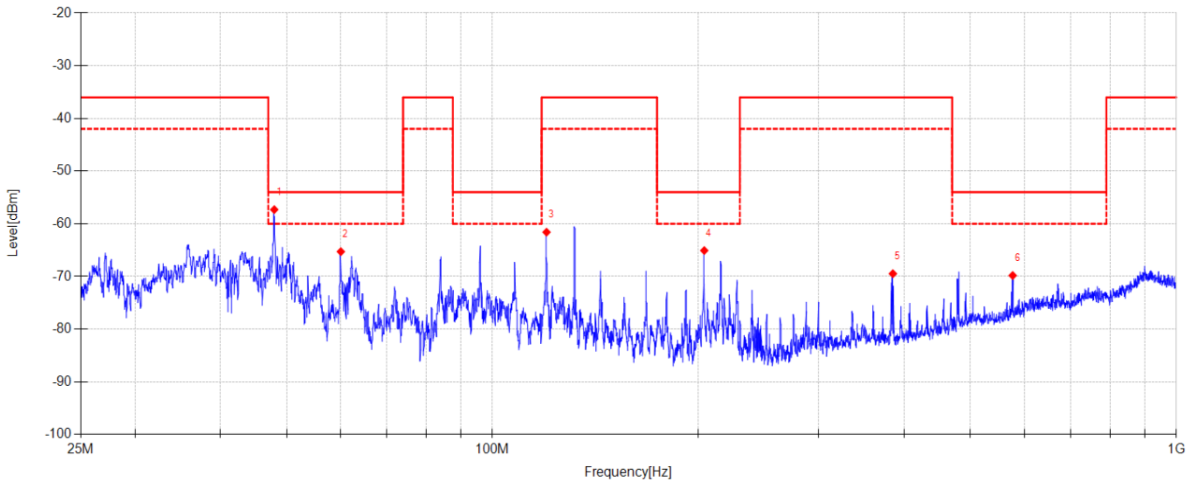



| Suspected Data List | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
| 1 | 30.018 | 51.22 | -114.30 | -63.08 | -36.00 | 27.08 | PK | Horizontal | ERP |
| 2 | 47.917 | 48.44 | -111.38 | -62.94 | -54.00 | 8.94 | PK | Horizontal | ERP |
| 3 | 119.950 | 41.20 | -117.45 | -76.25 | -36.00 | 40.25 | PK | Horizontal | ERP |
| 4 | 192.177 | 43.66 | -116.76 | -73.10 | -54.00 | 19.10 | PK | Horizontal | ERP |
| 5 | 335.896 | 46.71 | -110.67 | -63.96 | -36.00 | 27.96 | PK | Horizontal | ERP |
| 6 | 576.368 | 42.38 | -105.91 | -63.53 | -54.00 | 9.53 | PK | Horizontal | ERP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 100 kHz, VBW: 300 kHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-12-16 | **Tested By**: | Li Xiongbin |
| **Test Mode**: | TX 864.1MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Below 1G 868\34 | | |
| **Memo**: | Sample Number:S25091517-003 | | |




**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 47.952 | 58.79 | -116.10 | -57.31 | -54.00 | 3.31 | PK | Vertical | ERP |
| 2 | 60.006 | 50.55 | -115.84 | -65.29 | -54.00 | 11.29 | PK | Vertical | ERP |
| 3 | 119.950 | 51.03 | -112.62 | -61.59 | -36.00 | 25.59 | PK | Vertical | ERP |
| 4 | 204.009 | 49.43 | -114.51 | -65.08 | -54.00 | 11.08 | PK | Vertical | ERP |
| 5 | 385.008 | 38.47 | -107.95 | -69.48 | -36.00 | 33.48 | PK | Vertical | ERP |
| 6 | 576.368 | 33.98 | -103.80 | -69.82 | -54.00 | 15.82 | PK | Vertical | ERP |

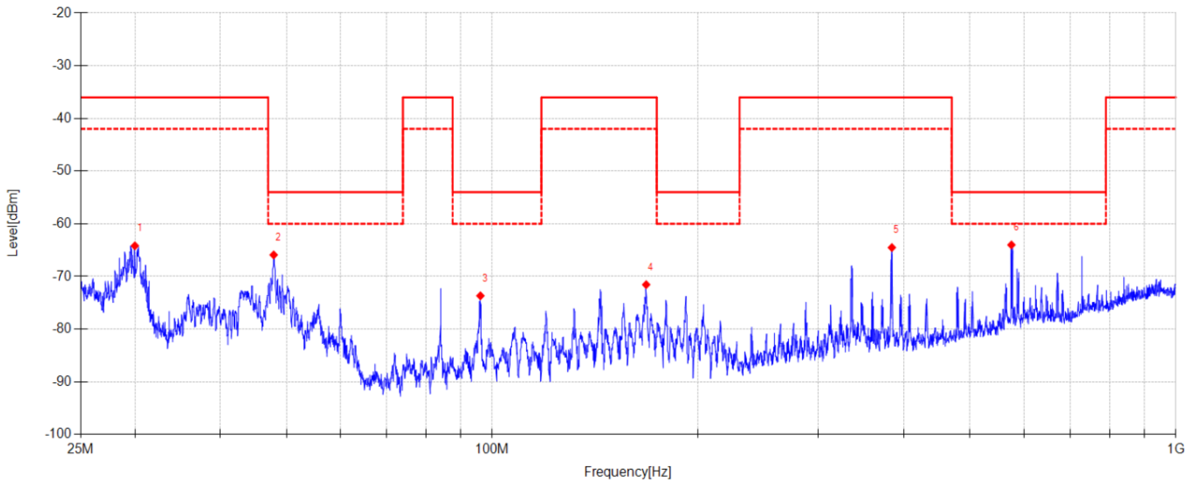
**Final Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 47.952 | 51.70 | -116.10 | -64.40 | -54.00 | 10.40 | RMS | Vertical | ERP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 100 kHz, VBW: 300 kHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-12-16 | **Tested By**: | Li Xiongbin |
| **Test Mode**: | TX 868.5MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Below 1G 868\35 | | |
| **Memo**: | Sample Number:S25091517-003 | | |



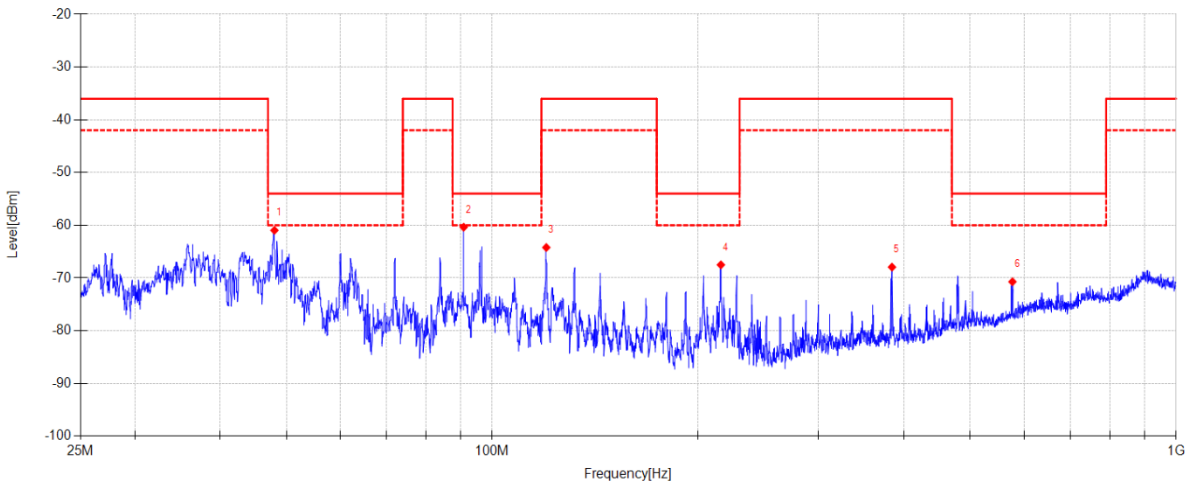

**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 29.996 | 50.12 | -114.30 | -64.18 | -36.00 | 28.18 | PK | Horizontal | ERP |
| 2 | 47.881 | 45.48 | -111.39 | -65.91 | -54.00 | 11.91 | PK | Horizontal | ERP |
| 3 | 96.067 | 43.17 | -116.85 | -73.68 | -54.00 | 19.68 | PK | Horizontal | ERP |
| 4 | 167.911 | 46.70 | -118.26 | -71.56 | -36.00 | 35.56 | PK | Horizontal | ERP |
| 5 | 384.157 | 45.15 | -109.67 | -64.52 | -36.00 | 28.52 | PK | Horizontal | ERP |
| 6 | 574.670 | 41.97 | -105.98 | -64.01 | -54.00 | 10.01 | PK | Horizontal | ERP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 100 kHz, VBW: 300 kHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-12-16 | **Tested By**: | Li Xiongbin |
| **Test Mode**: | TX 868.5MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Below 1G 868\36 | | |
| **Memo**: | Sample Number:S25091517-003 | | |




**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBµV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 48.023 | 55.13 | -116.09 | -60.96 | -54.00 | 6.96 | PK | Vertical | ERP |
| 2 | 90.830 | 52.55 | -112.92 | -60.37 | -54.00 | 6.37 | PK | Vertical | ERP |
| 3 | 119.950 | 48.40 | -112.62 | -64.22 | -36.00 | 28.22 | PK | Vertical | ERP |
| 4 | 215.932 | 46.69 | -114.20 | -67.51 | -54.00 | 13.51 | PK | Vertical | ERP |
| 5 | 384.157 | 40.01 | -107.95 | -67.94 | -36.00 | 31.94 | PK | Vertical | ERP |
| 6 | 576.368 | 33.08 | -103.80 | -70.72 | -54.00 | 16.72 | PK | Vertical | ERP |

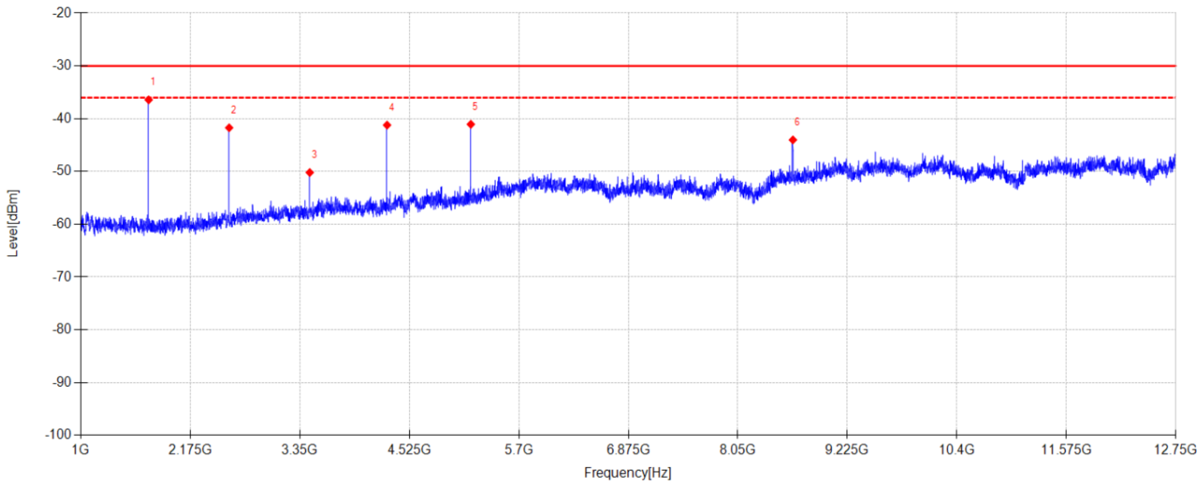
**Final Data List**

| NO. | Freq. [MHz] | Reading [dBµV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 90.830 | 37.07 | -112.92 | -75.85 | -54.00 | 21.85 | RMS | Vertical | ERP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 100 kHz, VBW: 300 kHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-12-16 | **Tested By**: | Li Xiongbin |
| **Test Mode**: | TX 864.1MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Above 1G 868\13 | | |
| **Memo**: | Sample Number:S25091517-003 | | |



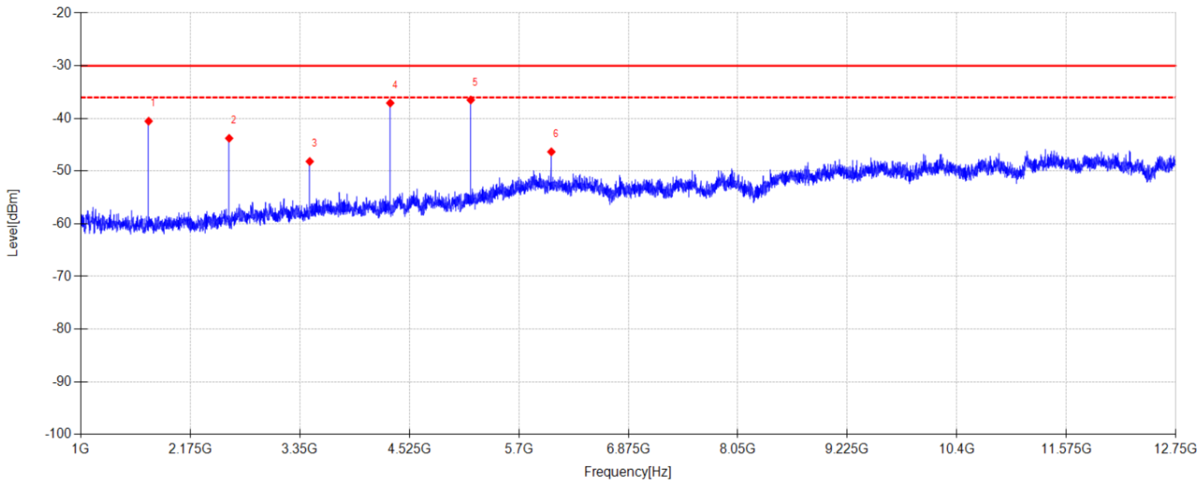

**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1728.500 | 68.20 | -104.63 | -36.43 | -30.00 | 6.43 | PK | Horizontal | EIRP |
| 2 | 2592.125 | 61.78 | -103.50 | -41.72 | -30.00 | 11.72 | PK | Horizontal | EIRP |
| 3 | 3456.925 | 51.76 | -101.96 | -50.20 | -30.00 | 20.20 | PK | Horizontal | EIRP |
| 4 | 4286.475 | 59.09 | -100.31 | -41.22 | -30.00 | 11.22 | PK | Horizontal | EIRP |
| 5 | 5185.350 | 56.49 | -97.55 | -41.06 | -30.00 | 11.06 | PK | Horizontal | EIRP |
| 6 | 8641.025 | 47.00 | -91.02 | -44.02 | -30.00 | 14.02 | PK | Horizontal | EIRP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss + Filter Factor - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-12-16 | **Tested By**: | Li Xiongbin |
| **Test Mode**: | TX 864.1MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Above 1G 868\14 | | |
| **Memo**: | Sample Number:S25091517-003 | | |

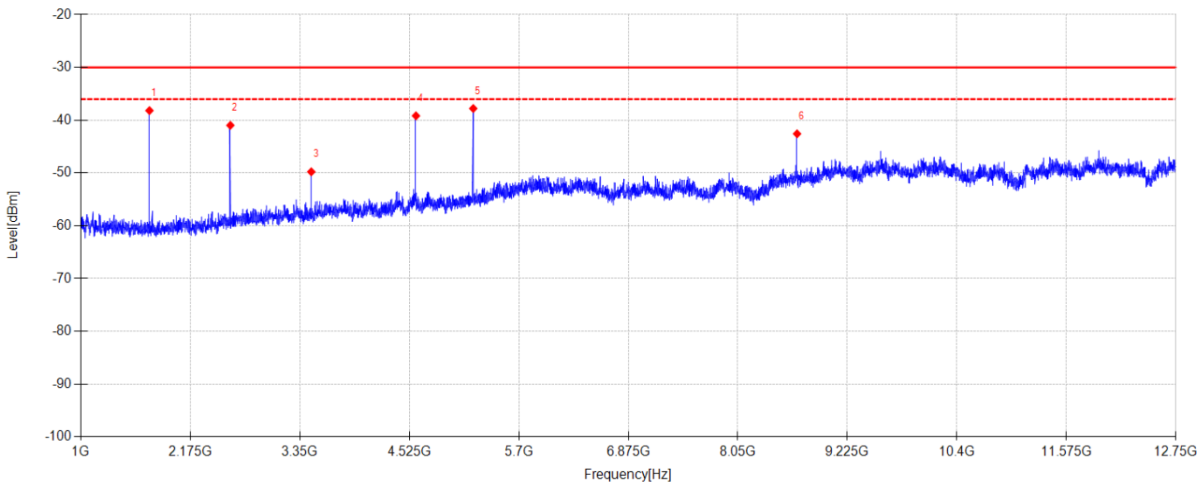



| Suspected Data List | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
| 1 | 1728.500 | 63.82 | -104.34 | -40.52 | -30.00 | 10.52 | PK | Vertical | EIRP |
| 2 | 2592.125 | 59.59 | -103.36 | -43.77 | -30.00 | 13.77 | PK | Vertical | EIRP |
| 3 | 3456.925 | 53.77 | -101.94 | -48.17 | -30.00 | 18.17 | PK | Vertical | EIRP |
| 4 | 4320.550 | 63.32 | -100.39 | -37.07 | -30.00 | 7.07 | PK | Vertical | EIRP |
| 5 | 5184.175 | 61.46 | -97.96 | -36.50 | -30.00 | 6.50 | PK | Vertical | EIRP |
| 6 | 6048.975 | 47.77 | -94.11 | -46.34 | -30.00 | 16.34 | PK | Vertical | EIRP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss + Filter Factor - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-12-16 | **Tested By**: | Li Xiongbin |
| **Test Mode**: | TX 868.5MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Above 1G 868\15 | | |
| **Memo**: | Sample Number:S25091517-003 | | |

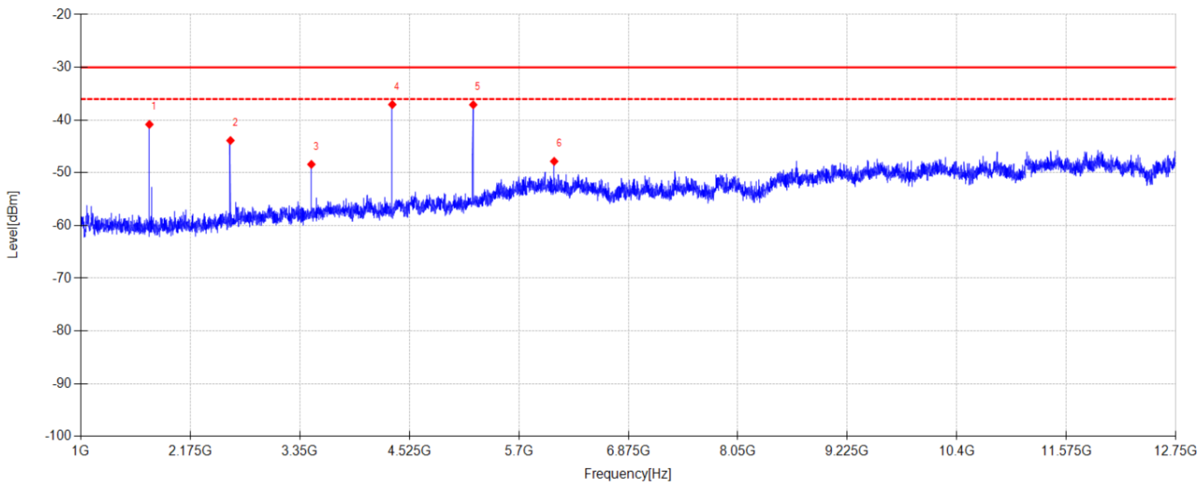



| Suspected Data List | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
| 1 | 1736.725 | 66.46 | -104.64 | -38.18 | -30.00 | 8.18 | PK | Horizontal | EIRP |
| 2 | 2605.050 | 62.51 | -103.49 | -40.98 | -30.00 | 10.98 | PK | Horizontal | EIRP |
| 3 | 3473.375 | 52.15 | -101.93 | -49.78 | -30.00 | 19.78 | PK | Horizontal | EIRP |
| 4 | 4596.675 | 60.28 | -99.45 | -39.17 | -30.00 | 9.17 | PK | Horizontal | EIRP |
| 5 | 5211.200 | 59.66 | -97.45 | -37.79 | -30.00 | 7.79 | PK | Horizontal | EIRP |
| 6 | 8685.675 | 48.27 | -90.85 | -42.58 | -30.00 | 12.58 | PK | Horizontal | EIRP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss + Filter Factor - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-12-16 | **Tested By**: | Li Xiongbin |
| **Test Mode**: | TX 868.5MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Above 1G 868\16 | | |
| **Memo**: | Sample Number:S25091517-003 | | |

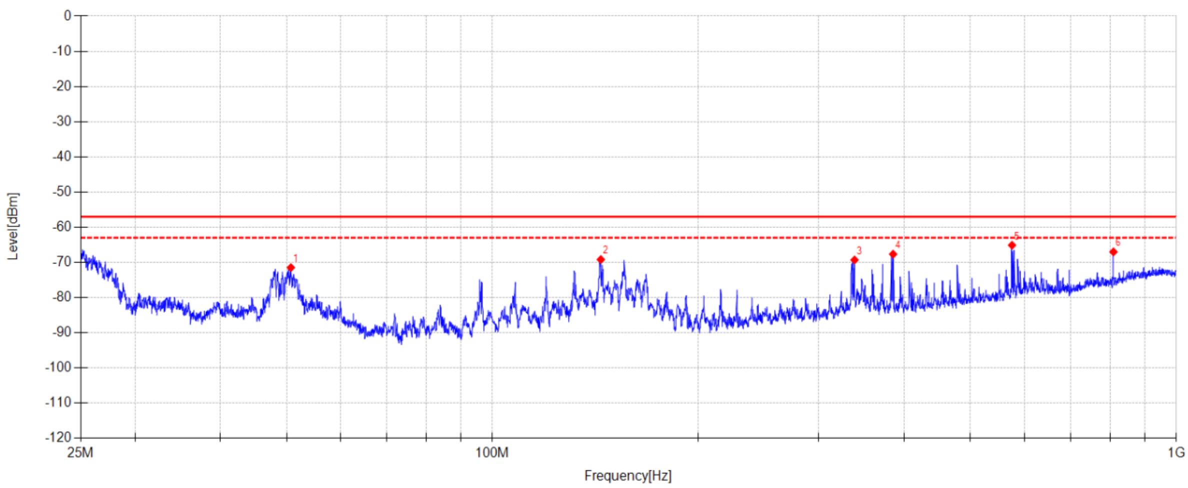



| Suspected Data List | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
| 1 | 1736.725 | 63.51 | -104.34 | -40.83 | -30.00 | 10.83 | PK | Vertical | EIRP |
| 2 | 2606.225 | 59.47 | -103.35 | -43.88 | -30.00 | 13.88 | PK | Vertical | EIRP |
| 3 | 3474.550 | 53.52 | -101.92 | -48.40 | -30.00 | 18.40 | PK | Vertical | EIRP |
| 4 | 4342.875 | 63.29 | -100.34 | -37.05 | -30.00 | 7.05 | PK | Vertical | EIRP |
| 5 | 5211.200 | 60.74 | -97.84 | -37.10 | -30.00 | 7.10 | PK | Vertical | EIRP |
| 6 | 6079.525 | 46.31 | -94.14 | -47.83 | -30.00 | 17.83 | PK | Vertical | EIRP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss + Filter Factor - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-11-26 | **Tested By**: | Li Xiongbin |
| **EUT**: | LoRa module | **Model Number**: | RFM90C |
| **Test Mode**: | RX 864.1MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Below 1G 868\9 | | |
| **Memo**: | Sample Numble:S25091517-003 | | |



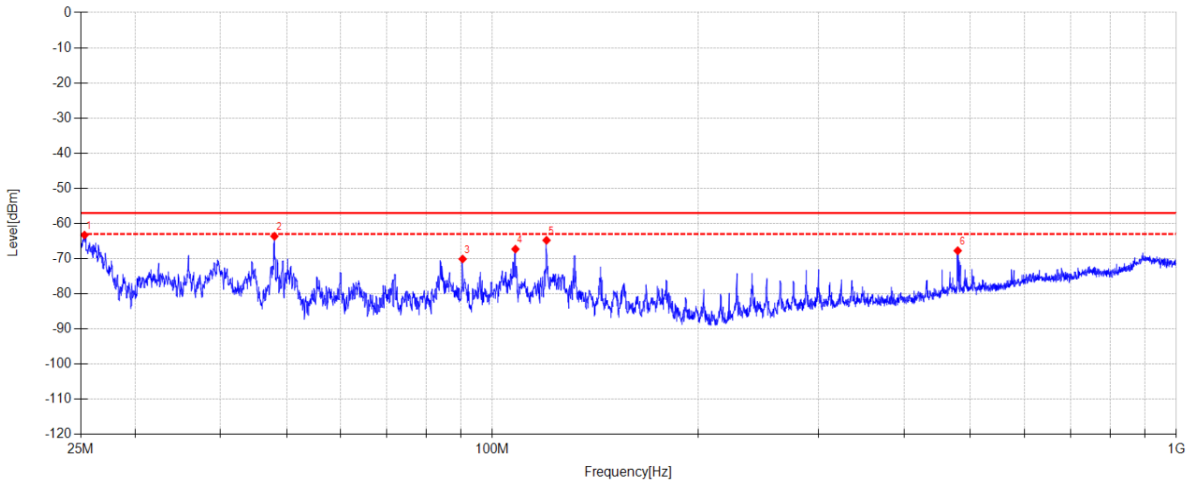

**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 50.717 | 39.81 | -111.29 | -71.48 | -57.00 | 14.48 | PK | Horizontal | ERP |
| 2 | 144.028 | 49.77 | -118.96 | -69.19 | -57.00 | 12.19 | PK | Horizontal | ERP |
| 3 | 338.383 | 41.28 | -110.58 | -69.30 | -57.00 | 12.30 | PK | Horizontal | ERP |
| 4 | 385.292 | 41.99 | -109.65 | -67.66 | -57.00 | 10.66 | PK | Horizontal | ERP |
| 5 | 575.094 | 40.84 | -105.96 | -65.12 | -57.00 | 8.12 | PK | Horizontal | ERP |
| 6 | 809.208 | 35.92 | -102.94 | -67.02 | -57.00 | 10.02 | PK | Horizontal | ERP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 100 kHz, VBW: 300 kHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-11-26 | **Tested By**: | Li Xiongbin |
| **EUT**: | LoRa module | **Model Number**: | RFM90C |
| **Test Mode**: | RX 864.1MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Below 1G 868\10 | | |
| **Memo**: | Sample Numble:S25091517-003 | | |

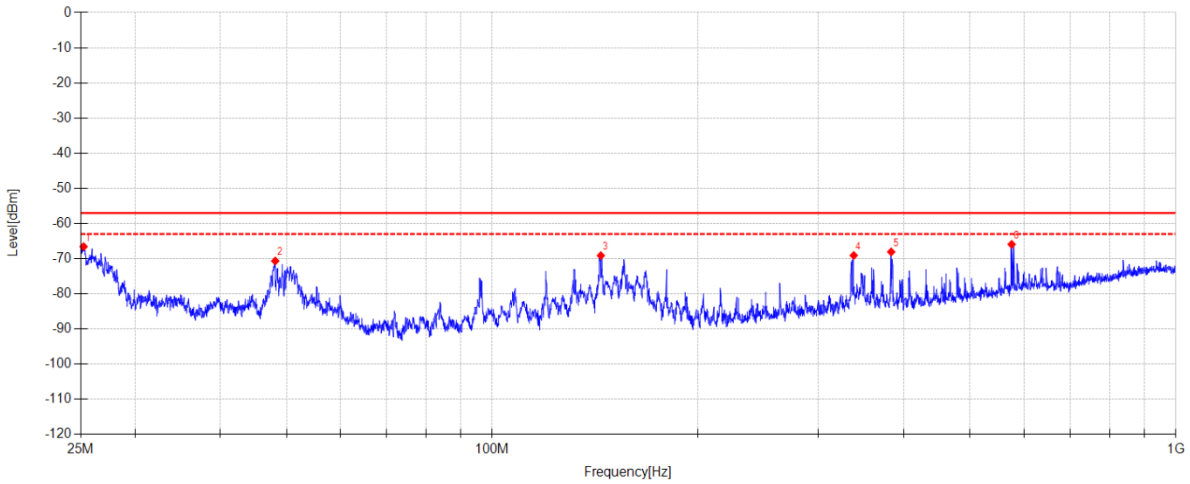



| Suspected Data List | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
| 1 | 25.315 | 49.99 | -113.29 | -63.30 | -57.00 | 6.30 | PK | Vertical | ERP |
| 2 | 47.987 | 52.47 | -116.10 | -63.63 | -57.00 | 6.63 | PK | Vertical | ERP |
| 3 | 90.429 | 42.88 | -112.99 | -70.11 | -57.00 | 13.11 | PK | Vertical | ERP |
| 4 | 108.022 | 44.50 | -111.78 | -67.28 | -57.00 | 10.28 | PK | Vertical | ERP |
| 5 | 119.950 | 47.84 | -112.62 | -64.78 | -57.00 | 7.78 | PK | Vertical | ERP |
| 6 | 478.953 | 37.99 | -105.74 | -67.75 | -57.00 | 10.75 | PK | Vertical | ERP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 100 kHz, VBW: 300 kHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-11-26 | **Tested By**: | Li Xiongbin |
| **EUT**: | LoRa module | **Model Number**: | RFM90C |
| **Test Mode**: | RX 868.5MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Below 1G 868\11 | | |
| **Memo**: | Sample Numble:S25091517-003 | | |



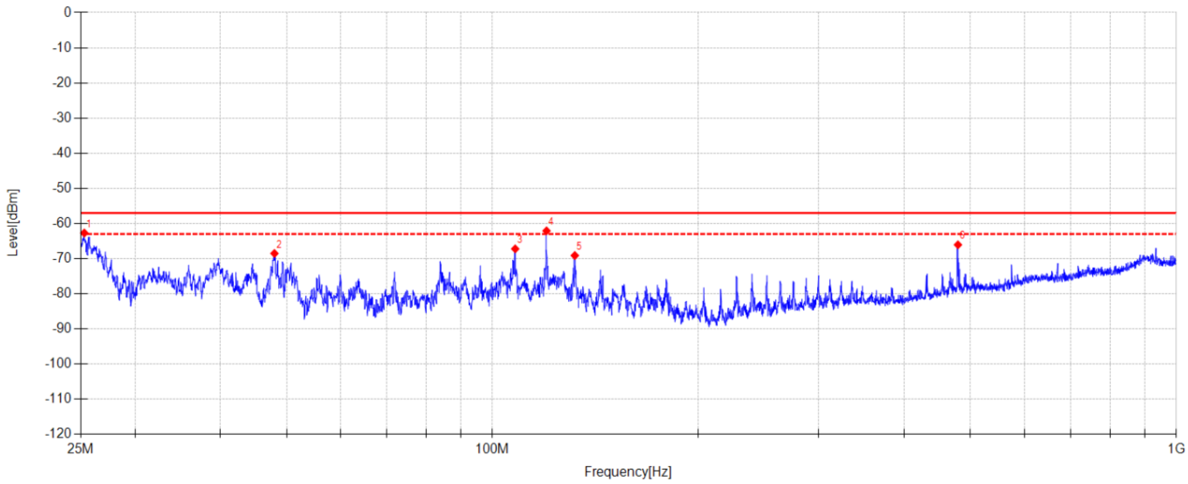

**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 25.241 | 43.58 | -110.16 | -66.58 | -57.00 | 9.58 | PK | Horizontal | ERP |
| 2 | 48.129 | 40.70 | -111.36 | -70.66 | -57.00 | 13.66 | PK | Horizontal | ERP |
| 3 | 144.134 | 49.84 | -118.97 | -69.13 | -57.00 | 12.13 | PK | Horizontal | ERP |
| 4 | 337.884 | 41.51 | -110.60 | -69.09 | -57.00 | 12.09 | PK | Horizontal | ERP |
| 5 | 383.308 | 41.59 | -109.68 | -68.09 | -57.00 | 11.09 | PK | Horizontal | ERP |
| 6 | 575.094 | 40.03 | -105.96 | -65.93 | -57.00 | 8.93 | PK | Horizontal | ERP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 100 kHz, VBW: 300 kHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-11-26 | **Tested By**: | Li Xiongbin |
| **EUT**: | LoRa module | **Model Number**: | RFM90C |
| **Test Mode**: | RX 868.5MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Below 1G 868\12 | | |
| **Memo**: | Sample Numble:S25091517-003 | | |



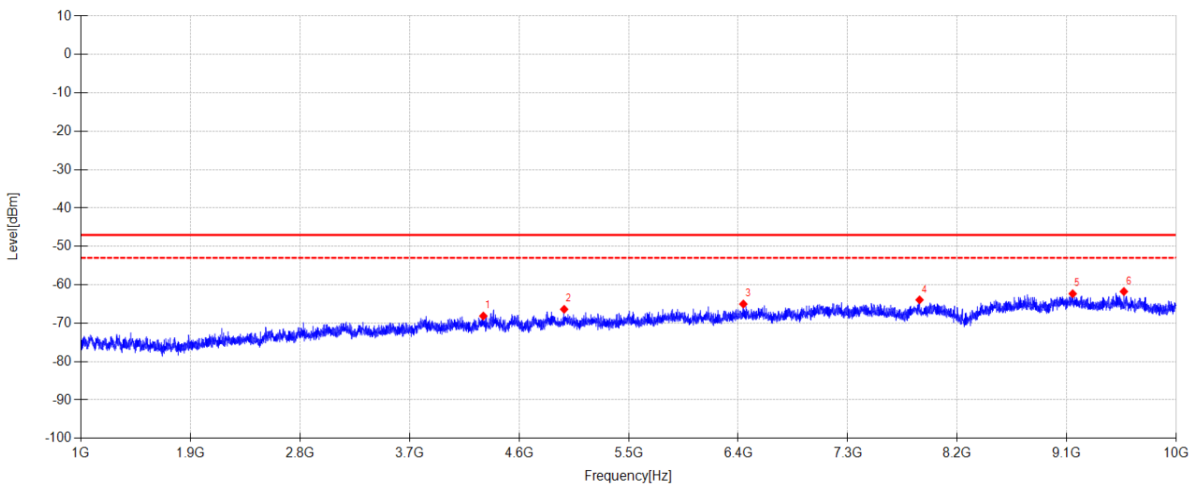

| Suspected Data List | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NO. | Freq. [MHz] | Reading [dBµV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
| 1 | 25.297 | 50.57 | -113.28 | -62.71 | -57.00 | 5.71 | PK | Vertical | ERP |
| 2 | 47.987 | 47.57 | -116.10 | -68.53 | -57.00 | 11.53 | PK | Vertical | ERP |
| 3 | 107.942 | 44.56 | -111.78 | -67.22 | -57.00 | 10.22 | PK | Vertical | ERP |
| 4 | 119.950 | 50.54 | -112.62 | -62.08 | -57.00 | 5.08 | PK | Vertical | ERP |
| 5 | 131.924 | 44.34 | -113.46 | -69.12 | -57.00 | 12.12 | PK | Vertical | ERP |
| 6 | 479.306 | 39.67 | -105.73 | -66.06 | -57.00 | 9.06 | PK | Vertical | ERP |

| Final Data List | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NO. | Freq. [MHz] | Reading [dBµV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
| 1 | 25.297 | 45.72 | -113.28 | -67.56 | -57.00 | 10.56 | RMS | Vertical | ERP |
| 2 | 119.950 | 45.39 | -112.62 | -67.23 | -57.00 | 10.23 | RMS | Vertical | ERP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 100 kHz, VBW: 300 kHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-11-26 | **Tested By**: | Li Xiongbin |
| **EUT**: | LoRa module | **Model Number**: | RFM90C |
| **Test Mode**: | RX 864.1MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Above 1G 868\5 | | |
| **Memo**: | Sample Numble:S25091517-003 | | |



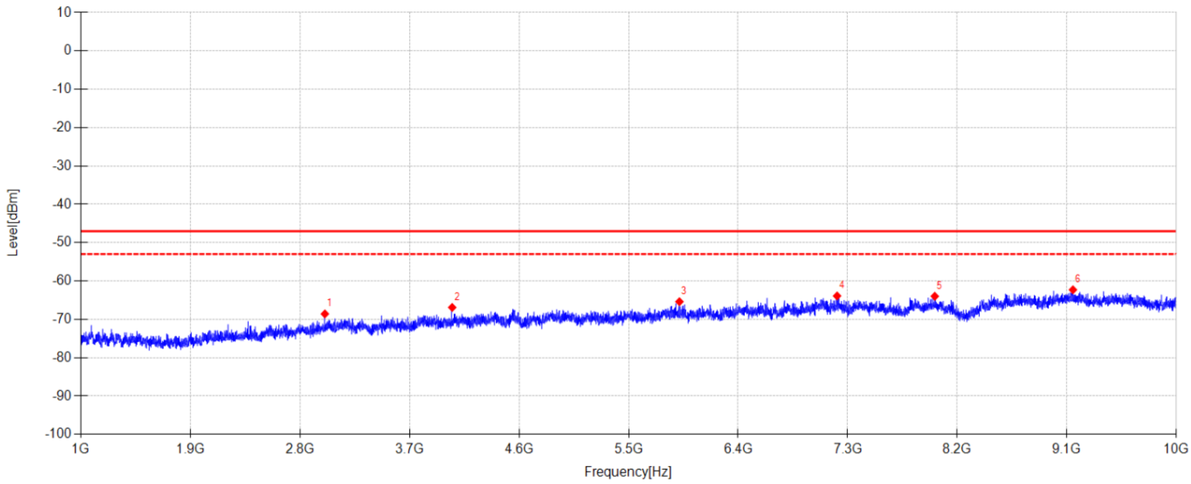

**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4306.600 | 42.11 | -110.25 | -68.14 | -47.00 | 21.14 | RMS | Horizontal | EIRP |
| 2 | 4970.800 | 42.41 | -108.79 | -66.38 | -47.00 | 19.38 | RMS | Horizontal | EIRP |
| 3 | 6443.200 | 41.17 | -106.19 | -65.02 | -47.00 | 18.02 | RMS | Horizontal | EIRP |
| 4 | 7890.400 | 40.48 | -104.41 | -63.93 | -47.00 | 16.93 | RMS | Horizontal | EIRP |
| 5 | 9148.600 | 38.23 | -100.56 | -62.33 | -47.00 | 15.33 | RMS | Horizontal | EIRP |
| 6 | 9568.900 | 40.02 | -101.77 | -61.75 | -47.00 | 14.75 | RMS | Horizontal | EIRP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss + Filter Factor - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-11-26 | **Tested By**: | Li Xiongbin |
| **EUT**: | LoRa module | **Model Number**: | RFM90C |
| **Test Mode**: | RX 864.1MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Above 1G 868\6 | | |
| **Memo**: | Sample Numble:S25091517-003 | | |



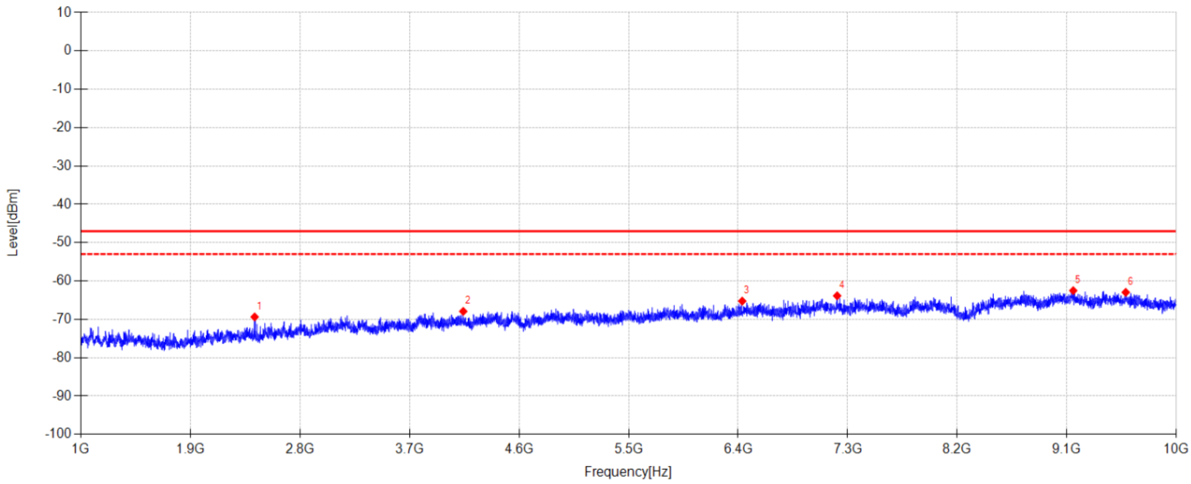

**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 3005.200 | 44.46 | -113.04 | -68.58 | -47.00 | 21.58 | RMS | Vertical | EIRP |
| 2 | 4051.000 | 44.18 | -111.08 | -66.90 | -47.00 | 19.90 | RMS | Vertical | EIRP |
| 3 | 5917.600 | 41.55 | -106.94 | -65.39 | -47.00 | 18.39 | RMS | Vertical | EIRP |
| 4 | 7212.700 | 41.04 | -104.93 | -63.89 | -47.00 | 16.89 | RMS | Vertical | EIRP |
| 5 | 8015.500 | 39.71 | -103.69 | -63.98 | -47.00 | 16.98 | RMS | Vertical | EIRP |
| 6 | 9150.400 | 38.24 | -100.54 | -62.30 | -47.00 | 15.30 | RMS | Vertical | EIRP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss + Filter Factor - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

| | | | |
|---|---|---|---|
| **Test Date**: | 2025-11-26 | **Tested By**: | Li Xiongbin |
| **EUT**: | LoRa module | **Model Number**: | RFM90C |
| **Test Mode**: | RX 868.5MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Above 1G 868\7 | | |
| **Memo**: | Sample Numble:S25091517-003 | | |



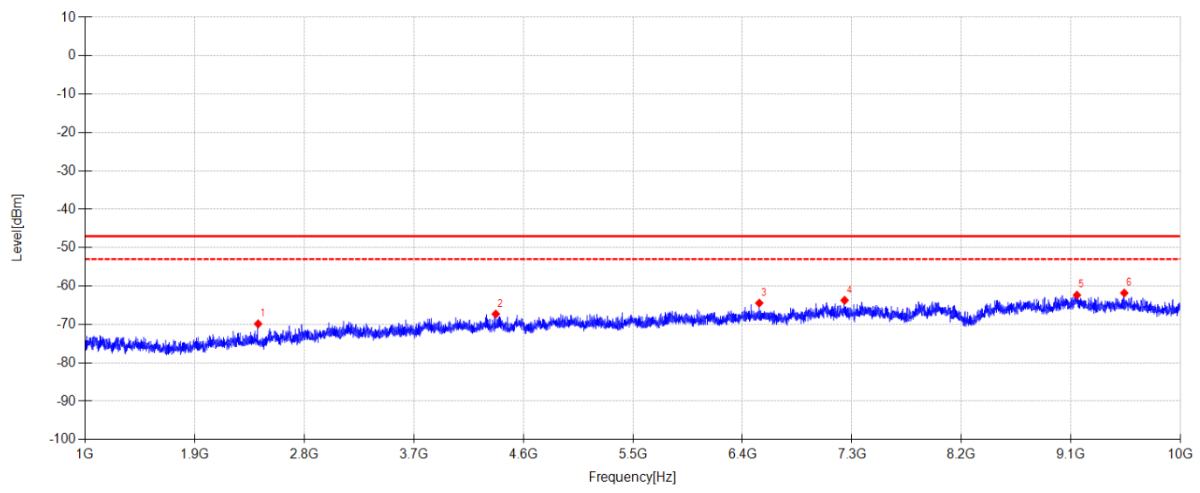

**Suspected Data List**

| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2428.300 | 45.93 | -115.30 | -69.37 | **-47.00** | 22.37 | RMS | Horizontal | EIRP |
| 2 | 4141.900 | 42.87 | -110.79 | -67.92 | **-47.00** | 20.92 | RMS | Horizontal | EIRP |
| 3 | 6433.300 | 40.97 | -106.21 | -65.24 | **-47.00** | 18.24 | RMS | Horizontal | EIRP |
| 4 | 7213.600 | 41.06 | -104.92 | -63.86 | **-47.00** | 16.86 | RMS | Horizontal | EIRP |
| 5 | 9153.100 | 38.08 | -100.58 | -62.50 | **-47.00** | 15.50 | RMS | Horizontal | EIRP |
| 6 | 9584.200 | 38.88 | -101.84 | -62.96 | **-47.00** | 15.96 | RMS | Horizontal | EIRP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss + Filter Factor - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.

# TR-4-E-009 Radiated Emission Test Result

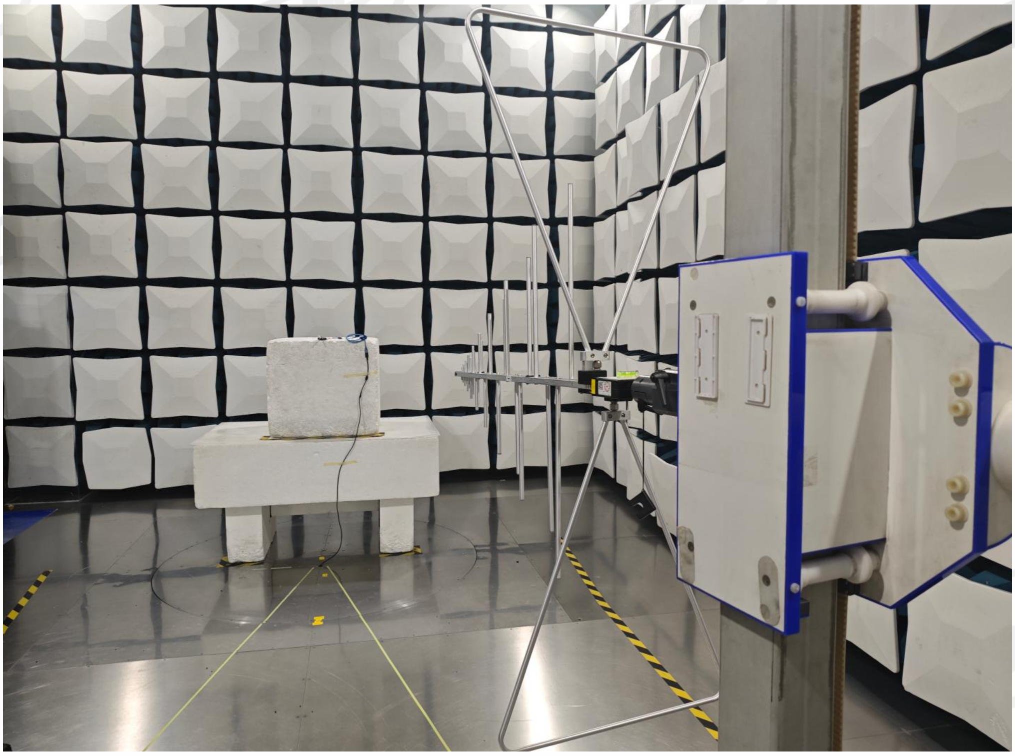
| | | | |
|---|---|---|---|
| **Test Date**: | 2025-11-26 | **Tested By**: | Li Xiongbin |
| **EUT**: | LoRa module | **Model Number**: | RFM90C |
| **Test Mode**: | RX 868.5MHz Mode | **Power Supply**: | DC 3.3V from control board |
| **Condition**: | Temp:22.8°C;Humi:43.4% | **Test Site**: | DDT 3# Chamber |
| **File Path**: | d:\ts\2025 report date\Q25091517-1E\CE Above 1G 868\8 | | |
| **Memo**: | Sample Numble:S25091517-003 | | |

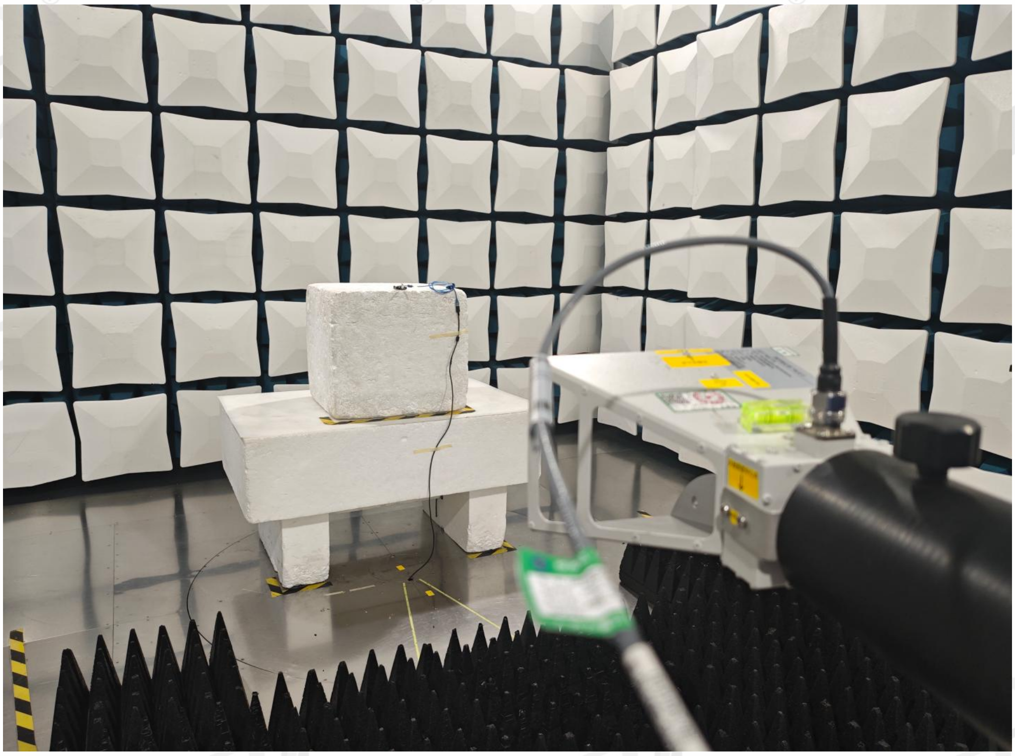



| Suspected Data List | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NO. | Freq. [MHz] | Reading [dBμV/m] | Factor [dB] | Level [dBm] | Limit [dBm] | Margin [dB] | Detector | Polarity | Type |
| 1 | 2422.000 | 45.42 | -115.28 | -69.86 | -47.00 | 22.86 | RMS | Vertical | EIRP |
| 2 | 4374.100 | 43.25 | -110.54 | -67.29 | -47.00 | 20.29 | RMS | Vertical | EIRP |
| 3 | 6538.600 | 41.75 | -106.19 | -64.44 | -47.00 | 17.44 | RMS | Vertical | EIRP |
| 4 | 7239.700 | 40.87 | -104.59 | -63.72 | -47.00 | 16.72 | RMS | Vertical | EIRP |
| 5 | 9148.600 | 38.20 | -100.56 | -62.36 | -47.00 | 15.36 | RMS | Vertical | EIRP |
| 6 | 9537.400 | 39.68 | -101.50 | -61.82 | -47.00 | 14.82 | RMS | Vertical | EIRP |

Note:
1. Level = Reading + Factor.
2. Factor = Antenna Factor + Cable Loss + Filter Factor - Preamp Gain + Site Loss Factor - 107.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.

# 16. Test Setup Photograph





# 17. Photos of the EUT

Please refer to appendix Ⅰ

-----------------------End Report----------------------